
March camp 2019 - Geometry

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1. PROBLEMS

1.1. Big picture!

Problem 1. Let ω be circumcircle of an acute triangle ABC . Point X lies inside ABC , such that

$$\sphericalangle BAX = 2\sphericalangle XBA \quad \text{and} \quad \sphericalangle XAC = 2\sphericalangle ACX.$$

M is middlepoint of arc BC of ω , which contains point A . Show that $XM = XA$.

Problem 2. Let the excircle of the triangle ABC lying opposite to A touch its side BC at the point A_1 . Define the points B_1 and C_1 analogously. Suppose that the circumcentre of the triangle $A_1B_1C_1$ lies on the circumcircle of the triangle ABC . Prove that the triangle ABC is right-angled.

Problem 3. Let I be the incenter of triangle ABC . Denote by $D, S \neq A$ intersections of AI with BC and circumcircle ω of ABC , respectively. Let K, L be incenters of triangles DSB and DCS . Let P be a reflection of I with respect to KL . Prove that $\sphericalangle BPC = 90^\circ$.

Problem 4. Given is triangle ABC . The angle bisectors of angles ABC and BCA intersect the sides CA and AB at the points B_1 and C_1 , and intersect each other at the point I . The line B_1C_1 intersects the circumcircle of triangle ABC at the points M and N . Prove that the circumradius of triangle MIN is twice as long as the circumradius of triangle ABC .

Problem 5. Points A, B lie on circle ω . Points C and D are moved on the arc AB , such that CD has constant length. Points I_1, I_2 are incenters of ABC and ABD , respectively. Prove that line I_1I_2 is tangent to some fixed circle.

Problem 6. Let ABC be a triangle with $\angle BAC = 60^\circ$. Let D and E be the feet of the perpendiculars from A to the external angle bisectors of ABC and ACB , respectively. Let O be the circumcenter of the triangle ABC . Prove that the circumcircles of the triangles ADE and BOC are tangent to each other.

Problem 7. Points L and K lie on BC and AC , respectively such that AL and BK are angle bisectors in the non-isosceles triangle ABC . The perpendicular bisector of BK intersects the line AL at point M . Point N lies on the line BK such that LN is parallel to MK . Prove that $LN = NA$.

Problem 8. Let ABC be a triangle, let D be the touchpoint of the side BC and the incircle of the triangle ABC , and let J_b and J_c be the incenters of the triangles ABD and ACD , respectively. Prove that the circumcentre of the triangle AJ_bJ_c lies on the bisectrix of the angle BAC .

Problem 9. Let K and L be two points on the arcs AB and BC of the circumcircle of a triangle ABC , respectively, such that $KL \parallel AC$. Show that the incenters of triangles ABK and CBL are equidistant from the midpoint of the arc AC , containing point B , of the circumcircle of triangle ABC .

Problem 10. Let AD be altitude in acute-angled triangle ABC . Points M and N are projections of point D onto AB and AC . Lines MN and AD intersect circumcircle ω of triangle ABC respectively at points P , Q and A , R . Prove that D is incenter of PQR .

Problem 11. Given is a triangle ABC . The points A , B , C divide the circumcircle Ω of the triangle ABC into three arcs BC , CA , AB . Let X be a variable point on the minor arc AB , and let O_1 and O_2 be the incenters of the triangles CAX and CBX . Prove that the circumcircle of the triangle XO_1O_2 intersects the circle Ω in a fixed point.

Problem 12. Given is triangle ABC ($AB > BC$) inscribed in the circle Ω . On the sides AB and BC , points M and N are chosen, respectively, so that $AM = CN$. Lines MN and AC intersect at point K . Let P be the center of the inscribed circle of triangle AMK , and Q the center of the K -excircle of the triangle CNK . Prove that the midpoint of the arc AC , containing point B , of the circle Ω is equidistant from the points P and Q .

Problem 13. Denote by I the incenter of ABC and by I_A the A -excenter. Let M be the midpoint of segment CB and let N be the midpoint of arc BC , which contains point A . T is symmetric point to the point N in point reflection with respect to A . Prove that the quadrilateral $I_A MIT$ is cyclic.

Problem 14. The circle Γ has centre O , and BC is a diameter of Γ . Let A be a point which lies on Γ such that $\angle AOB < 120^\circ$. Let D be the midpoint of the arc AB which does not contain C . The line through O parallel to DA meets the line AC at I . The perpendicular bisector of OA meets Γ at E and at F . Prove that I is the incentre of the triangle CEF .

Problem 15. Let H be the orthocenter of the triangle ABC . Let M and N be the midpoints of the sides AB and AC , respectively. Assume that H lies inside the quadrilateral $BMNC$ and that the circumcircles of triangles BMH and CNH are tangent to each other. The line through H parallel to BC intersects the circumcircles of the triangles BMH and CNH in the points K and L , respectively. Let F be the intersection point of MK and NL and let J be the incenter of triangle MHN . Prove that $FJ = FA$.

Problem 16. A trapezoid $ABCD$ in which $AB \parallel CD$ and $AB > CD$, is circumscribed. The incircle of the triangle ABC touches the lines AB and AC at the points M and N , respectively. Prove that the incenter of the trapezoid $ABCD$ lies on the line MN .

Problem 17. Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. The inscribed circle ω of triangle BCD meets CD at E . Let F be a point on the (internal) angle bisector of $\angle DAC$ such that $EF \perp CD$. Let the circumscribed circle of triangle ACF meet line CD at C and G . Prove that the triangle AFG is isosceles.

Problem 18. Let N be the midpoint of arc AC containing point B of the circumcircle of triangle ABC , let M be the midpoint of segment AC and let I_1, I_2 be the incentres of triangles ABM and CBM . Prove that quadrilateral points $I_1 I_2 BN$ is cyclic.

Problem 19. Let $ABCD$ be a convex quadrilateral with $BA \neq BC$. Denote the incircles of triangles ABC and ADC by ω_1 and ω_2 respectively. Suppose that there exists a circle ω tangent to ray BA beyond A and to the ray BC beyond C , which is also tangent to the lines AD and CD . Prove that the common external tangents to ω_1 and ω_2 intersect on ω .

Problem 20. Let I be the incentre of a non-equilateral triangle ABC , I_A be the A -excentre, I'_A be the reflection of I_A in BC , and l_A be the reflection of line AI'_A in AI . Define points I_B , I'_B and line l_B analogously. Let P be the intersection point of l_A and l_B .

(a) Prove that P lies on line OI where O is the circumcentre of triangle ABC .

(b) Let one of the tangents from P to the incircle of triangle ABC meet the circumcircle at points X and Y . Show that $\angle X I Y = 120^\circ$.

1.2. Power of a point.

Problem 1. Given is triangle ABC . Points P and Q were chosen such that $\angle PBC = \angle QCB = 90^\circ$ and $AP = PB$, $AQ = CQ$.

Tangent to circumcircle of ABC passing through point A intersects BC at R . Prove that P, Q and R are collinear.

Problem 2. Let CX, CY be the tangents from vertex C of triangle ABC to the circle passing through the midpoints of its sides. Prove that lines XY , AB and the tangent at point C to the circumcircle ω of ABC concur at one point.

Problem 3. An acute triangle ABC in which $AB < AC$ is given. Points E and F are feet of its heights from B and C , respectively. The line tangent in point A to the circle circumscribed on ABC crosses BC at P . The line parallel to BC that goes through point A crosses EF at Q . Prove PQ is perpendicular to the median from A of triangle ABC .

Problem 4. Let circles ω_1 and ω_2 , with centres in O_1, O_2 , respectively, intersect at two distinct points P and Q . Their common tangent, closer to point P , touches the circles at A, B respectively. Let the perpendicular from A to the line BP meet O_1O_2 at C . Prove that $\angle APC = 90^\circ$.

Problem 5. The circumradius and the inradius of triangle ABC are equal to R and r ; O, I are the centers of respective circles. External bisector of angle C intersect AB in point P . Point Q is the projection of P to line OI . Find distance OQ .

Problem 6. A circle, its chord AB and the midpoint W of the minor arc AB are given. Take an arbitrary point C on the major arc AB . The tangent to the circle at C meets the tangents at A and B at points X and Y , respectively. Lines WX and WY meet AB at points N and M respectively. Prove that the length of segment NM does not depend on point C .

Problem 7. The diagonals of a cyclic quadrilateral $ABCD$ meet at point M . A circle ω touches segments MA and MD at points P, Q respectively and is tangent to the circumcircle of $ABCD$ at point X . Prove that X lies on the radical axis of circumcircles of ACQ and BDP .

Problem 8. Let L and K be the feet of the internal and the external bisectors of angle BAC in a triangle ABC . Let P be the common point of the tangents to the circumcircle of the triangle at B and C . The perpendicular from L to BC meets AP at point Q . Prove that Q lies on the medial line of triangle LKP .

Problem 9. Lines b and c passing through vertices B and C of triangle ABC are perpendicular to sideline BC . The perpendicular bisectors to AC and AB meet b and c at points P and Q , respectively. Prove that the line PQ is perpendicular to median AM of triangle ABC .

Problem 10. In triangle ABC , let ω be the excircle opposite to A . Let D, E and F be the points where ω is tangent to BC, CA , and AB , respectively. The circle AEF intersects line BC at P and Q . Let M be the midpoint of AD . Prove that the circle MPQ is tangent to ω .

Problem 11. Let ABC be a triangle with $\angle BCA = 90^\circ$, and let D be the foot of the altitude from C . Let X be a point in the interior of the segment CD . Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. Let M be the point of intersection of AL and BK . Show that $MK = ML$.

Problem 12. Let ABC be a triangle. The incircle of ABC touches the sides AB and AC at the points Z and Y , respectively. Let G be the point where the lines BY and CZ meet, and let R and S be points such that the two quadrilaterals $BCYR$ and $BCSZ$ are parallelogram. Prove that $GR = GS$.

Problem 13. Let $ABCD$ be a convex quadrilateral whose sides AD and BC are not parallel. Suppose that the circles with diameters AB and CD meet at points E and F inside the quadrilateral. Let ω_E be the circle through the feet of the perpendiculars from E to the lines AB, BC and CD . Let ω_F be the circle through the feet of the perpendiculars from F to the lines CD, DA and AB . Prove that the midpoint of the segment EF lies on the line through the two intersections of ω_E and ω_F .

Problem 14. Let $ABCD$ be a circumscribed quadrilateral. Its incircle ω touches sides BC and DA at points E and F respectively. It is known that lines AB, FE and CD concur. The circumcircles of triangles AED and BFC meet ω for the second time at points E_1 and F_1 . Prove that $EF \parallel E_1F_1$.

Problem 15. Circles ω_B and ω_C are excircles of a triangle ABC . The circle ω'_B is symmetric to ω_B with respect to the midpoint of AC , the circle ω'_C is symmetric to ω_C with respect to the midpoint of AB . Prove that the radical axis of ω'_B and ω'_C halves the perimeter of ABC .

1.3. SSS - Similarity, (Spiral) Similarity & Symmedians.

Problem 1. Let ABC be an acute triangle in which $AB \neq AC$. Point J is an A -excentre of triangle ABC ; A -excircle touches segment BC at E . Lines BC and AJ intersect at D . Circumcircles of triangles ABC and ADE intersect at $F \neq A$. Prove that $\angle AFJ = 90^\circ$.

Problem 2. Perpendicular bisector of side BC intersects circumcircle of triangle ABC in points P and Q . Points A and P lie on the same side of line BC . Point R is an orthogonal projection of point P on line AC . Point S is middle of line segment AQ . Show that points A, B, R, S lie on one circle.

Problem 3. Let ABC be a triangle with $AB = AC \neq BC$ and let I be its incentre. The line BI meets AC at D , and the line through D perpendicular to AC meets AI at E . Prove that the reflection of I in AC lies on the circumcircle of triangle BDE .

Problem 4. Consider the parallelogram $ABCD$ with obtuse angle at A . Let H be the feet of perpendicular from A to the side BC . The median from C in triangle ABC meets the circumcircle of triangle ABC at the point K . Prove that points $KHCD$ is cyclic.

Problem 5. Let $ABCD$ be a quadrilateral. Let diagonals AC and BD meet at P . Let O_1 and O_2 be the circumcenters of APD and BPC . Let M, N , and O be the midpoints of AC, BD , and O_1O_2 . Show that O is the circumcenter of MPN .

Problem 6. In triangle ABC denote by K the midpoint of BC . Let M be a point of the segment AB . Line KM intersects AC at L , such that C lies between A and L . Point N is a midpoint of LM . Line AN intersects circumcircle of ABC at $S \neq A$. Prove that if $S \neq N$, then circle passing through K, N and S is tangent to BC .

Problem 7. Two circles intersect at two points A and B . A line ℓ which passes through the point A meets the two circles again at the points C and D , respectively. Let M and N be the midpoints of the arcs BC and BD (which do not contain the point A) on the respective circles. Let K be the midpoint of the segment CD . Prove that $\angle MKN = 90^\circ$.

Problem 8. Let $ABCD$ be a convex quadrilateral. The perpendicular bisectors of its sides AB and CD meet at Y . Denote by X a point inside the quadrilateral $ABCD$ such that $\angle ADX = \angle BCX < 90^\circ$ and $\angle DAX = \angle CBX < 90^\circ$. Show that $\angle AYB = 2 \cdot \angle ADX$.

Proof. <https://artofproblemsolving.com/community/c6h59215p360389> $\square$

Problem 9. In triangle ABC point M is a midpoint of AB . Point D lies inside this triangle and satisfies conditions

$$\angle DAC = \angle ABC \quad \text{and} \quad \angle DCA = \angle BCM.$$

Show that $DM \parallel BC$.

Problem 1. Let $ABCDE$ be a convex pentagon such that

$$\angle BAC = \angle CAD = \angle DAE \quad \text{and} \quad \angle ABC = \angle ACD = \angle ADE.$$

The diagonals BD and CE meet at P . Prove that the line AP bisects the side CD .

Problem 2. Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC , respectively, such that $\frac{AE}{ED} = \frac{BF}{FC}$. Ray FE meets rays BA and CD at S and T , respectively. Prove that the circumcircles of triangles SAE , SBF , TCF , and TDE pass through a common point.

Problem 3. Let ABC be a triangle and let M and N denote the midpoints of $\overline{AB}$ and $\overline{AC}$, respectively. Let X be a point such that $\overline{AX}$ is tangent to the circumcircle of triangle ABC . Denote by ω_B the circle through M and B tangent to $\overline{MX}$, and by ω_C the circle through N and C tangent to $\overline{NX}$. Show that ω_B and ω_C intersect on line BC .

Proof. <https://artofproblemsolving.com/community/q2h1751587p11419585> $\square$

Problem 4. In acute angled $\triangle ABC$, $AB > AC$, points E, F lie on AC, AB respectively, satisfying $BF + CE = BC$. Let I_B, I_C be the excenters of $\triangle ABC$ opposite B, C respectively, EI_C, FI_B intersect at T , and let K be the midpoint of arc BAC . Let KT intersect the circumcircle of $\triangle ABC$ at K, P . Show T, F, P, E concyclic.

Proof. <https://artofproblemsolving.com/community/c6h1691830p10822375> $\square$

Problem 5. Let $ABCD$ be a cyclic quadrilateral with $AB < CD$. The diagonals intersect at the point F and lines AD and BC intersect at the point E . Let K and L be the orthogonal projections of F onto lines AD and BC respectively, and let M , S and T be the midpoints of EF , CF and DF respectively. Prove that the second intersection point of the circumcircles of triangles MKT and MLS lies on the segment CD .

Proof. <https://artofproblemsolving.com/community/c6h1239090p6316969> $\square$

1.4. Inversion $\sqrt{bc}$.

Problem 1. (Inversion $(\sqrt{bc})$) Given is a triangle ABC . Consider such transformation ϕ which is composition of inversion with respect to circle with center A and radius $\sqrt{AB \cdot AC}$ and symmetry about the angle bisector of angle BAC . Show that

- (a) $\phi(B) = C, \phi(C) = B$;
- (b) $\phi(\omega) = BC, \phi(BC) = \omega$, where ω is circumcircle of triangle ABC .

Problem 2. Given is a triangle ABC . Consider such transformation ϕ which is composition of inversion with respect to circle with center A and radius $\sqrt{\frac{1}{2}AB \cdot AC}$ and symmetry about the angle bisector of angle BAC . Prove

- (a) $\phi(O) = H_A$, where O is circumcenter of ABC , and H_A is the base of altitude from vertex A ;
- (b) circumcircle of triangle BOC is mapped to nine point circle of ABC .

Problem 3. Let Ω be circumcircle of triangle ABC . Denote by o circle tangent to AB, AC and internally tangent to ω at point T . Point D is tangency point of A -excircle with line BC . Prove that $\angle BAT = \angle DAC$.

Problem 4. Denote by O and I circumcenter and incenter of triangle ABC , respectively. Let circle tangent to AB, AC and circumcircle of triangle BOC be tangent to AB and AC at points M and N , respectively. Show that MN bisects AI .

Problem 5. Let ABC be a triangle and A', B', C' the symmetric of vertex about opposite sides. The intersection of the circumcircles of triangles ABB' and ACC' is A_1 . Points B_1 and C_1 are defined similarly. Prove that lines AA_1, BB_1 and CC_1 are concurrent.

Problem 6. Circles ω_1 and ω_2 intersect at X and Y and are tangent internally to circle Ω , respectively at points A and B . Line AB intersect again circles ω_1 and ω_2 at A_1, B_1 , respectively. Circle inscribed in curvilinear triangle A_1B_1X touches A_1B_1 at the point Z . Prove that $\angle AXZ = \angle BXZ$.

Problem 7. Consider point D on circumcircle of ABC such that AD is diameter of this circle. Let DB, DC intersect AC, AB at X, Y , respectively. ω_1 is circle passing through A, B and tangent to BC ; analogously ω_2 passes through A, C and is tangent to BC . Let ω_1 and ω_2 intersect at G ; circumcircle of AXY intersect ω_1, ω_2 at P, Q , respectively. Prove that circumcircle of triangle GPQ is tangents to BC .

Problem 8. Given is triangle ABC . Points D i E lie on line AB in order D, A, B, E . Moreover $AD = AC$ and $BE = BC$. Angle bisector of angles at BAC and ABC intersect BC, AC at P and Q , and circumcircle of ABC at M and N , respectively. Line connecting A with circumcenter of BME and line connecting B with circumcenter of AND intersect at X . Prove that $CX \perp PQ$.

1.5. Monge's theorem.

Problem 6. Let ABC be a triangle and Ω its circumcircle. Let Γ_A be the circle inside Ω tangent to Ω , AB and AC . Define similarly Γ_B and Γ_C . Let T_A be the point where Γ_A touches Ω , define similarly T_B and T_C . Prove that AT_A , BT_B and CT_C are concurrent.

Proof. <https://artofproblemsolving.com/community/c6h1556771p11708603> $\square$

Problem 7. Point P lies on side AB of a convex quadrilateral $ABCD$. Let ω be the incircle of triangle CPD , and let I be its incenter. Suppose that ω is tangent to the incircles of triangles APD and BPC at points K and L , respectively. Let lines AC and BD meet at E , and let lines AK and BL meet at F . Prove that points E , I , and F are collinear.

Proof. <https://artofproblemsolving.com/community/c6h214649p1186805> $\square$

Problem 8. Triangle ABC with $AB = AC$ is inscribed in circle o . Circles o_1 and o_2 are internally tangent to circle o in points P and Q , respectively, and they are tangent to segments AB and AC , respectively, and they are disjoint with the interior of triangle ABC . Let m be a line tangent to circles o_1 and o_2 , such that points P and Q lie on the opposite side than point A . Line m cuts segments AB and AC in points K and L , respectively. Prove, that intersection point of lines PK and QL lies on bisector of angle BAC .

Proof. <https://artofproblemsolving.com/community/c6h1632349p10251557> $\square$

2. SOLUTIONS TO PROBLEMS

2.1. Incenter, mid-arc.

Problem 1. Let ω be circumcircle of an acute triangle ABC . Point X lies inside ABC , such that $\angle BAX = 2\angle XBA$ and $\angle XAC = 2\angle ACX$. M is midpoint of arc BC of ω , which contains point A . Show that $XM = XA$.

Proof. Without loss of generality we can assume that $AB < AC$ (if $AB = AC$, then $A \equiv M$ and the proof is trivial). Take E on ray $\overrightarrow{AB}$, that $AX = EX$. Since

$$\angle XEA = \angle BAX = 2\angle XBA > \angle XBA,$$

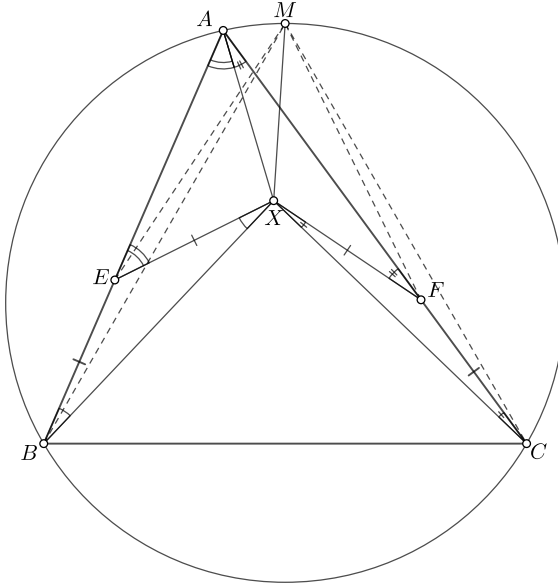
we see that E lies on segment AB . Moreover

$$2\angle XBE = 2\angle XBA = \angle XEA = 180^\circ - \angle BEX = \angle XBE + \angle EXB,$$

thus $\angle XBE = \angle EXB$, which means that EXB is an isosceles triangle and $EB = EX$.

Similarly, let F be a point on ray $\overrightarrow{AC}$ for which $FX = AX$. Then analogously we infer that F lies on segment AC and $FC = FX$. Therefore

$$BE = EX = AX = XF = FC.$$



Since M is a midpoint of arc BAC , then $BM = MC$. Moreover $\angle MBE = \angle MCF$. This equality, together with $BE = CF$, imply that triangles MBE and MCF are congruent. In particular $\angle EMB = \angle FMC$, thus

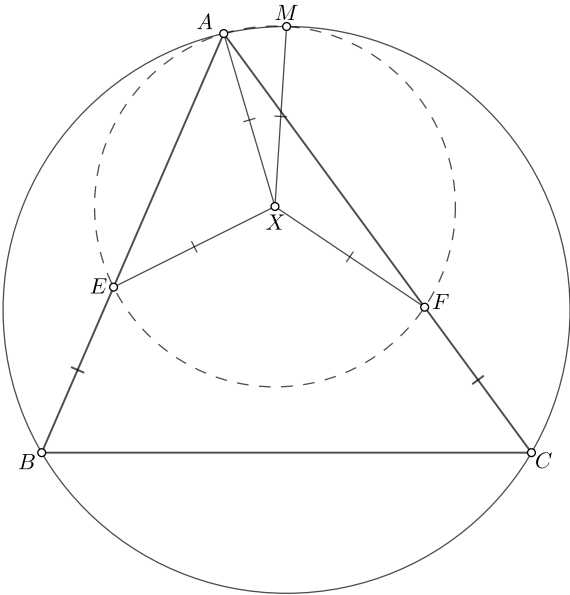
$$\angle EMF = \angle EMB + \angle BMF = \angle FMC + \angle BMF = \angle BMC.$$

The angles BAC and BMC are equal, because they are based on arc BC , thus

$$\angle EMF = \angle BMC = \angle BAC = \angle EAF,$$

which means that A, M, F and E lie on one circle.

The equality $BE = EX = AX = XF = FC$. proves that X is circumcenter of AFE , but the above consideration shows that point M is the centre of these circle, so $AX = XM$.



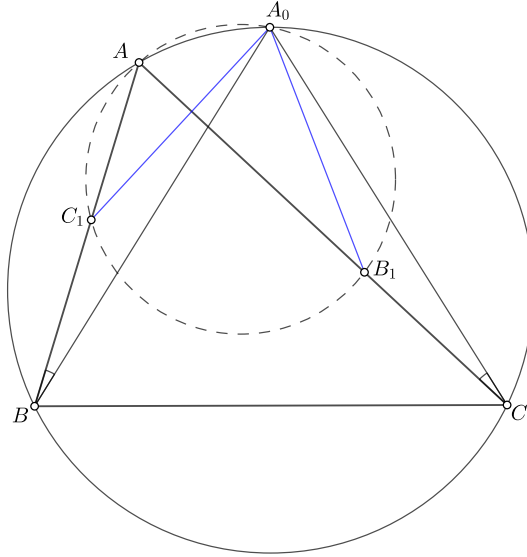
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Problem 2. Let the excircle of the triangle ABC lying opposite to A touch its side BC at the point A_1 . Define the points B_1 and C_1 analogously. Suppose that the circumcentre of the triangle $A_1B_1C_1$ lies on the circumcircle of the triangle ABC . Prove that the triangle ABC is right-angled.

Proof. Denote the circumcircles of the triangles ABC and $A_1B_1C_1$ by Ω and Γ , respectively. Denote the midpoint of the arc CB of Ω containing A by A_0 , and define B_0 as well as C_0 analogously. By our hypothesis the centre Q of Γ lies on Ω .

Lemma 1. One has $A_0B_1 = A_0C_1$. Moreover, the points A , A_0 , B_1 , and C_1 are concyclic. Finally, the points A and A_0 lie on the same side of B_1C_1 . Similar statements hold for B and C .

Proof of lemma. Let us consider the case $A = A_0$ first. Then the triangle ABC is isosceles at A , which implies $AB_1 = AC_1$ while the remaining assertions of the Lemma are obvious. So let us suppose $A \neq A_0$ from now on.



By the definition of A_0 , we have $A_0B = A_0C$. It is also well known and easy to show that $BC_1 = CB_1$. Next, we have

$$\sphericalangle C_1BA_0 = \sphericalangle ABA_0 = \sphericalangle ACA_0 = \sphericalangle B_1CA_0.$$

Hence the triangles A_0BC_1 and A_0CB_1 are congruent. This implies $A_0C_1 = A_0B_1$, establishing the first part of the Lemma. It also follows that $\sphericalangle A_0C_1A = \sphericalangle A_0B_1A$, as these are exterior angles at the corresponding vertices C_1 and B_1 of the congruent triangles A_0BC_1 and A_0CB_1 . For that reason the points A , A_0 , B_1 , and C_1 are indeed the vertices of some cyclic quadrilateral two opposite sides of which are AA_0 and B_1C_1 .

Now we turn to the solution. Evidently the points A_1 , B_1 , and C_1 lie interior to some semicircle arc of Γ , so the triangle $A_1B_1C_1$ is obtuse-angled. Without loss

of generality, we will assume that its angle at B_1 is obtuse. Thus Q and B_1 lie on different sides of A_1C_1 ; obviously, the same holds for the points B and B_1 . So, the points Q and B are on the same side of A_1C_1 .

Notice that the perpendicular bisector of A_1C_1 intersects Ω at two points lying on different sides of A_1C_1 . By the first statement from the Lemma, both points B_0 and Q are among these points of intersection; since they share the same side of A_1C_1 , they coincide.

Now, by the first part of the Lemma again, the lines QA_0 and QC_0 are the perpendicular bisectors of B_1C_1 and A_1B_1 , respectively. Thus

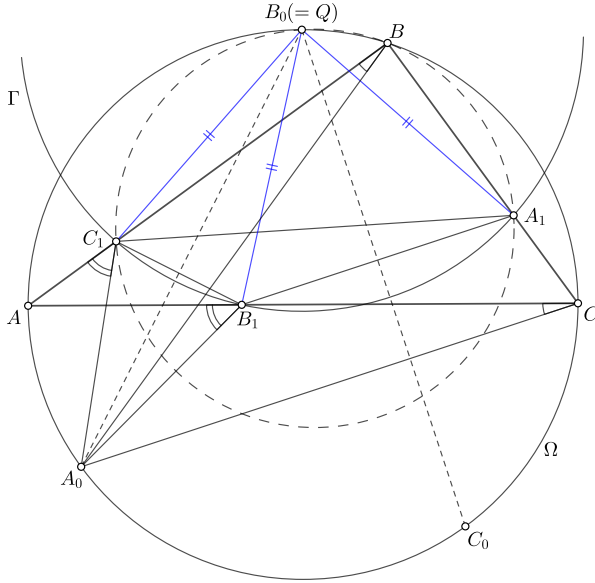
$$\begin{aligned} \sphericalangle C_1B_0A_1 &= \sphericalangle C_1B_0B_1 + \sphericalangle B_1B_0A_1 = 2\sphericalangle A_0B_0B_1 + 2\sphericalangle B_1B_0C_0 = \\ &= 2\sphericalangle A_0B_0C_0 = 180^\circ - \sphericalangle ABC, \end{aligned}$$

recalling that A_0 and C_0 are the midpoints of the arcs CB and BA , respectively.

On the other hand, by the second part of the Lemma we have

$$\sphericalangle C_1B_0A_1 = \sphericalangle C_1BA_1 = \sphericalangle ABC.$$

From the last two equalities, we get $\sphericalangle ABC = 90^\circ$, whereby the problem is solved.

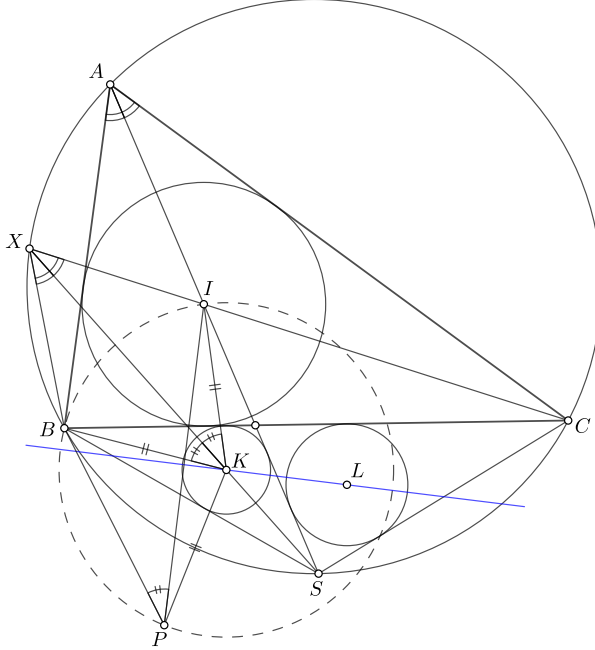


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Problem 3. Let I be the incenter of $\triangle ABC$. Denote by $D, S \neq A$ intersections of AI with BC and circumcircle ω of ABC , respectively. Let K, L be incenters of triangles DSB and DCS . Let P be a reflection of I with respect to KL . Prove that $BPC = 90^\circ$.

Proof. Denote by $X \neq S$ second intersection of SK with circumcircle of triangle ABC . Since X is midpoint of arc AB (arc not containing point C), thus angle bisector of $\angle ACB$ intersect with ω at X . Moreover S is midpoint of arc BC . Thereby

$$\angle BXS = \angle BAS = \angle SAC = \angle SXC \quad \text{and} \quad \angle ISX = \angle XSB.$$



We may now conclude that points I and B are symmetric with respect to line SX . In particular $KI = KB$. Since P and I are symmetric with respect to KL , hence

$$KP = KI = KB.$$

It means that points I, B, P lie on circle centred at K , so

$$\begin{aligned} \angle IPB &= \frac{1}{2} \angle IKB = \angle XKB = 180^\circ - \angle BKS = \\ &= \angle SBK + \angle KSB = \frac{1}{2} (\angle SBD + \angle DSB). \end{aligned}$$

Analogously we show that $\angle CPI = \frac{1}{2} (\angle CSD + \angle DCS)$. Eventually we obtain that

$$\begin{aligned}
\angle CPB &= \angle IPB + \angle CPI = \\
&= \frac{1}{2} (\angle SBD + \angle DSB + \angle CSD + \angle DCS) = \frac{1}{2} \cdot 180^\circ = 90^\circ.
\end{aligned}$$

□

Problem 4. Given is triangle ABC . The angle bisectors of angles ABC and BCA intersect the sides CA and AB at the points B_1 and C_1 , and intersect each other at the point I . The line B_1C_1 intersects the circumcircle of triangle ABC at the points M and N . Prove that the circumradius of triangle MIN is twice as long as the circumradius of triangle ABC .

Proof. Denote by I_b, I_c the B and, respectively, C -excenters of ABC . Recall these two well-known lemmas.

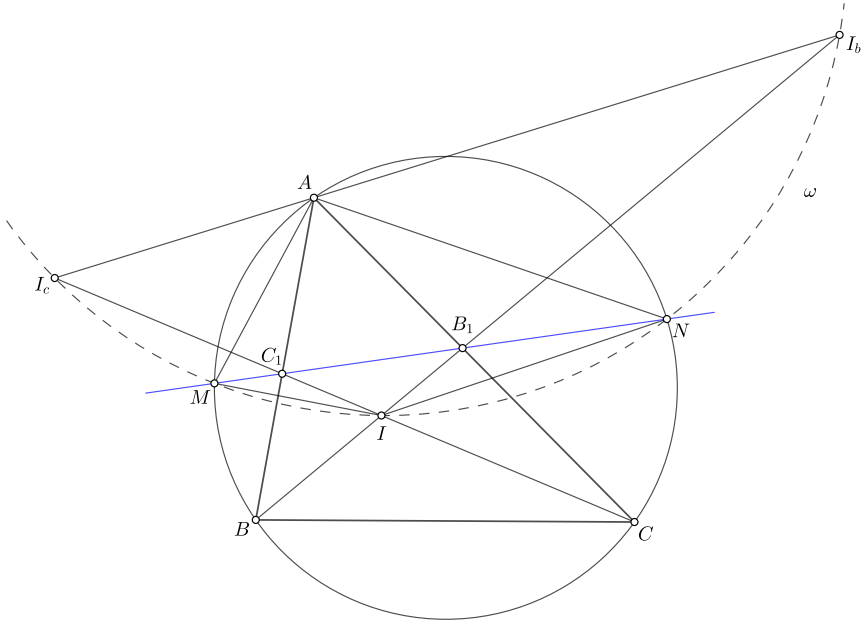
Lemma 1. Circumcircle of triangle ABC is nine point circle of triangle I_bII_c . It means that circumradius of II_bI_c is twice as long as the circumradius of triangle ABC .

Proof of lemma. One just need to apply trillium theorem.

Lemma 2. Let DEF be orthic triangle of ABC . Then the orthic axis¹ of ABC coincides with the radical axis of its circumcircle and nine-point circle.

Proof of lemma. The proof is trivial and left to the reader (one just need to apply Ceva's theorem and simple similarities of triangles).

If we manage to prove that the circumcircle ω of triangle MIN passes through I_b and I_c we will be done, because then ω would be the circumcircle of II_bI_c , while according to lemma 1 we know that circumradius of II_bI_c is twice as long as the circumradius of triangle ABC .



¹If $D \in BC$, $E \in CA$ and $F \in AB$ then the **orthic axis** of triangle ABC passes through points $X = AB \cap DE$, $Y = BC \cap EF$ and $Z = CA \cap FD$.

Notice that ABC is the orthic triangle of II_bI_c , so B_1C_1 is the orthic axis of II_bI_c , which coincides with the radical axis of its circumcircle and nine-point circle. These two circles do intersect, because $\angle I_bII_c > \frac{\pi}{2}$, so the two points of intersection must be precisely the points of intersection between B_1C_1 and (ABC) , i.e. M and N .

□

Problem 5. Points A, B lie on circle ω . Points C and D are moved on the arc AB , such that CD has constant length. Points I_1, I_2 are incenters of ABC and ABD , respectively. Prove that line I_1I_2 is tangent to some fixed circle.

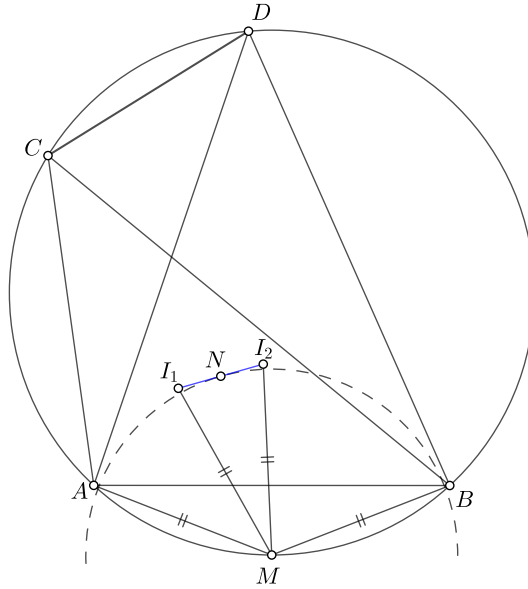
Proof. Denote by M the midpoint of arc AB not containing C and D . Observe that

$$\angle AI_1B = 90^\circ + \frac{1}{2}\angle ACB = 90^\circ + \frac{1}{2}\angle ADB = \angle AI_2B.$$

Thus AI_1I_2B is cyclic. Note that M lies on CI_1 and DI_2 . Since CD has constant length, I_1I_2 also has constant length. By the *incenter/excenter lemma*, we have

$$MA = MI_1 = MI_2 = MB.$$

Let N be the midpoint of I_1I_2 . Then $MN \perp I_1I_2$. Let ω be the circle with center M and radius MN . Notice that I_1I_2 is tangent to ω . As C varies over arc AB , the length MN remains constant. Thus ω is a fixed circle which gives us our desired result.



□

Problem 6. Let ABC be a triangle with $\angle BAC = 60^\circ$. Let D and E be the feet of the perpendiculars from A to the external angle bisectors of ABC and ACB , respectively. Let O be the circumcenter of the triangle ABC . Prove that the circumcircles of the triangles ADE and BOC are tangent to each other.

Proof. Denote by I_A the A -excenter of triangle ABC and by I incenter. We will show that circumcircles of the triangles ADE and BOC are tangent to each other at I_A .

Note that quadrilateral $BIOC$ is cyclic, since

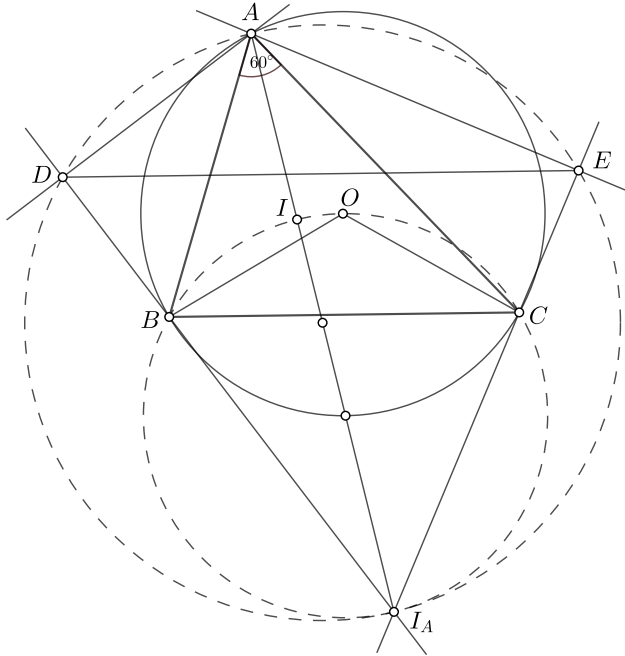
$$\angle BOC = 2 \cdot 60^\circ = 120^\circ$$

and

$$\angle BIC = 90^\circ + \frac{1}{2}\angle BAC = 120^\circ.$$

Moreover quadrilateral $BICI_A$ is cyclic from obvious reasons. Thus points B, I, O, C and I_A lie on one circle. Of course $I_A = DB \cap EC$ and $\angle ADI_A = \angle AEI_A = 90^\circ$. Hence quadrilateral $DAEI_A$ is cyclic too.

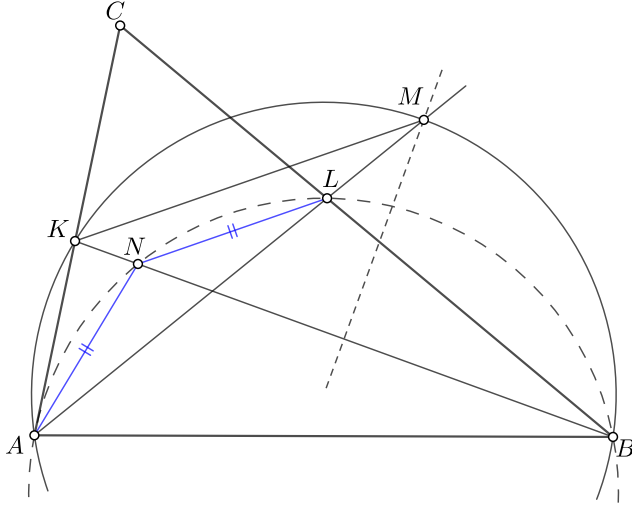
The centers of these circles are obviously the mid-points of AI_A, II_A , respectively. Since A, I, I_A are collinear, it means that circumcircles of $BIOCI_A$ and $DAEI_A$ are tangent at I_A .



□

Problem 7. Points L and K lie on BC and AC , respectively such that AL and BK are angle bisectors in the non-isosceles triangle ABC . The perpendicular bisector of BK intersects the line AL at point M . Point N lies on the line BK such that LN is parallel to MK . Prove that $LN = NA$.

Proof. Firstly, we check the case if $AB = AK$. Then AM coincides with perpendicular bisector of BK , so $BK \perp AM$. Moreover, because BK is bisector and altitude in $\triangle ABL$, it's perpendicular bisector of AL . Thus, $NL = NA$.



Now we consider case when $AB \neq AK$. Let circumcircle of ABK meet AM at M' . Then $KM' = M'B$, meaning that, if M and M' are distinct, MM' is perpendicular bisector of BK . It means that $AK = AB$, a contradiction. Thus, $M = M'$ and quadrilateral $BMKA$ is cyclic. Then

$$\sphericalangle KBA = \sphericalangle KMA = \sphericalangle NLA.$$

Hence quadrilateral $BLNA$ is also cyclic, and because BN is angle bisector, we finally obtain that $LN = NA$. $\square$

Problem 8. Let ABC be a triangle, let D be the touchpoint of the side BC and the incircle of the triangle ABC , and let J_b and J_c be the incentres of the triangles ABD and ACD , respectively. Prove that the circumcentre of the triangle AJ_bJ_c lies on the bisectrix of the angle BAC .

Proof. Let the incircle of the triangle ABC meet CA and AB at points E and F , respectively. Let the incircles of the triangles ABD and ACD meet AD at points X and Y , respectively. Then

$$2DX = DA + DB - AB = DA + DB - BF - AF = DA - AF;$$

similarly, $2DY = DA - AE = 2DX$. Hence the points X and Y coincide, so $J_bJ_c \perp AD$.

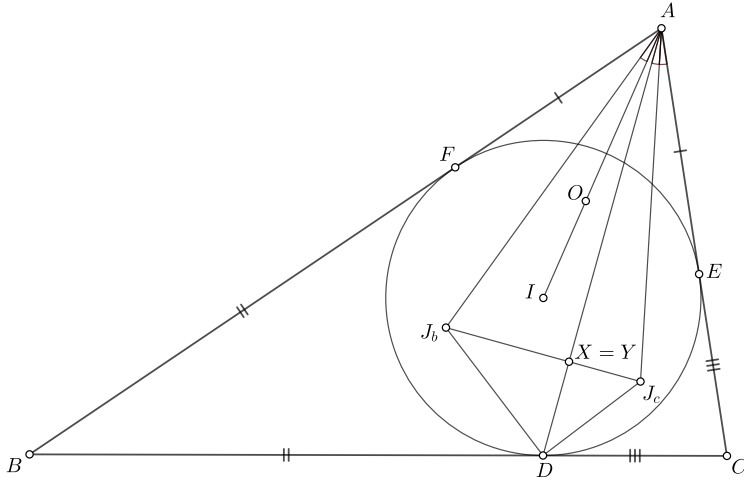
Now let O be the circumcentre of the triangle AJ_bJ_c . Then

$$\sphericalangle J_bAO = 90^\circ - \frac{1}{2}\sphericalangle AOJ_b = 90^\circ - \sphericalangle AJ_cJ_b = \sphericalangle XAJ_c = \frac{1}{2}\sphericalangle DAC.$$

Therefore

$$\sphericalangle BAO = \sphericalangle BAJ_b + \sphericalangle J_bAO = \frac{1}{2}\sphericalangle BAD + \frac{1}{2}\sphericalangle DAC = \frac{1}{2}\sphericalangle BAC,$$

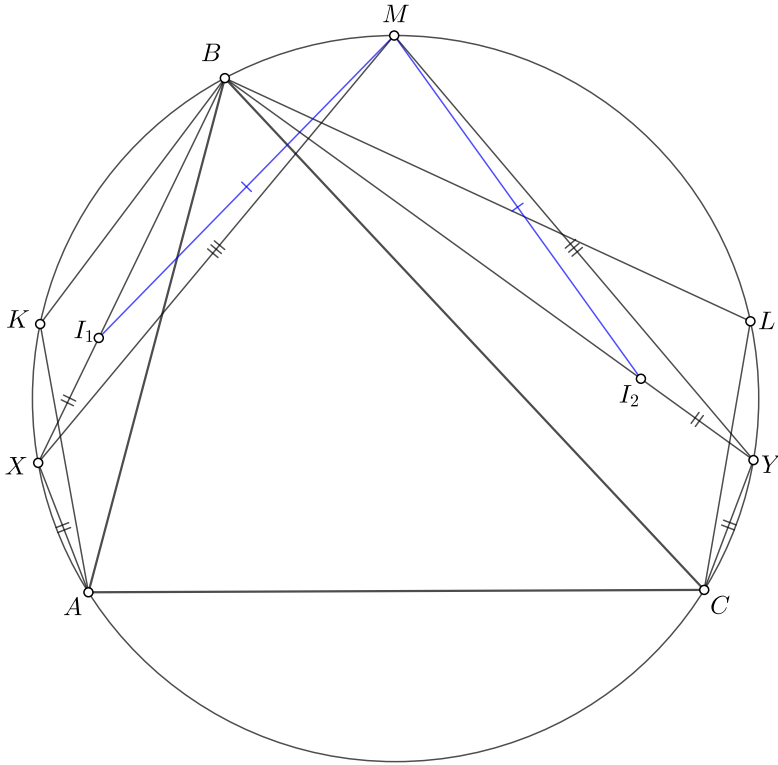
and the conclusion follows.



□

Problem 9. Let K and L be two points on the arcs AB and BC of the circumcircle of a triangle ABC , respectively, such that $KL \parallel AC$. Show that the incenters of triangles ABK and CBL are equidistant from the midpoint of the arc AC , containing point B , of the circumcircle of triangle ABC .

Proof. Let I_1 and I_2 be the incenters of triangles BKA and BLC , respectively. Intersect BI_1 and BI_2 with the circumcircle and denote those points of intersection as X and Y , respectively. Notice that $XACY$ is a trapezoid (because points X and Y are midpoints of minor arcs AK and BL ; moreover $KL \parallel AB$) so $XA = YC$. By trillium theorem $AX = XI_1$. Analogously $YC = YI_2$ so we conclude that $XI_1 = YI_2$. Notice that $MX = MY$, because M is the midpoint of arc AC and $KL \parallel AB$. Since $MX = MY$ and $\angle BXM = \angle BYM$ we can obtain that $\triangle MI_1X = \triangle MI_2Y$. It means that $MI_1 = MI_2$.



□

Problem 10. Let AD be altitude in acute-angled triangle ABC . Points M and N are projections of point D onto AB and AC . Lines MN and AD intersect circumcircle ω of triangle ABC respectively at points P , Q and A , R . Prove that D is incenter of PQR .

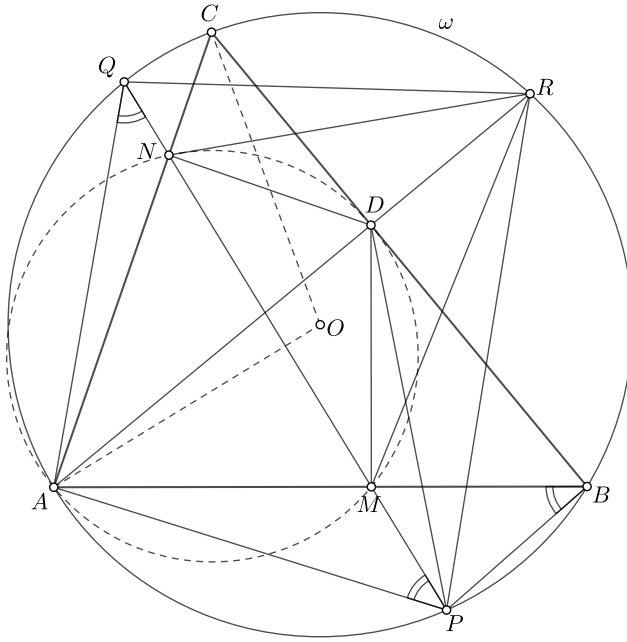
Proof. Firstly, we will show that point A is midpoint of arc PQ of ω . Circle with diameter AD contains points M and N , hence

$$\angle ANM = \angle ADM = 90^\circ - \angle DAB = \angle ABC.$$

Denote by O circumcenter of ω . Then

$$\angle NAO = \frac{1}{2} (180^\circ - \angle AOC) = \frac{1}{2} (180^\circ - 2\angle ABC) = 90^\circ - \angle ABC.$$

Thus $\angle ANM + \angle NAO = 90^\circ$. It means that $AO \perp MN$, hence A is indeed midpoint of arc PQ . In particular it means that D lies on angle bisector of PRQ .



Now we will show that D also lies on perpendicular bisector of angle RPQ . Since PAQ is an isosceles triangle, so

$$\angle APM = \angle AQP = \angle ABP.$$

Therefore triangles AMP and APB are similar. Right-angled triangles AMD and ADB are also similar. We can now conclude that

$$AP^2 = AM \cdot AB = AD^2.$$

Hence triangle PAD is isosceles too. It means that

$$\begin{aligned}\angle DPQ &= \angle APD - \angle APQ = \frac{1}{2} (180^\circ - \angle PAD) = \frac{1}{2} (180^\circ - \angle PAQ) = \\ &= \frac{1}{2} (\angle PAQ - \angle PAD) = \frac{1}{2} \angle RAQ = \frac{1}{2} \angle RPQ.\end{aligned}$$

Point D lies on angle bisector of angle RPQ . Hence D is incenter of triangle PRQ and we are done.

□

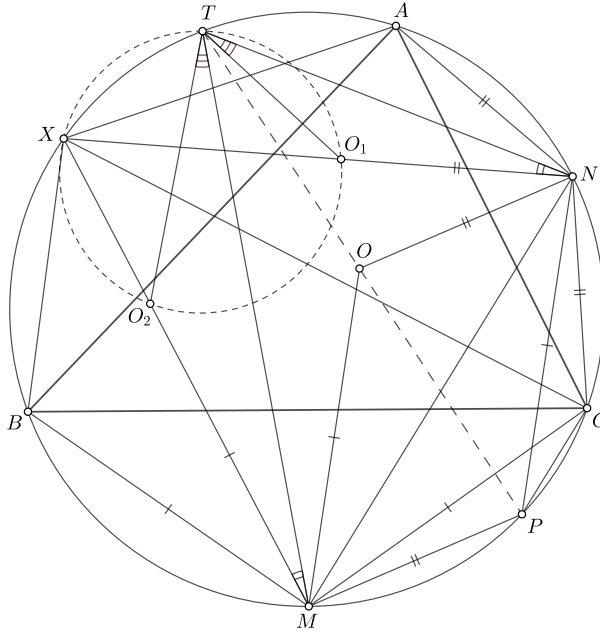
Problem 11. Given is a triangle ABC . The points A, B, C divide the circumcircle Ω of the triangle ABC into three arcs BC, CA, AB . Let X be a variable point on the minor arc AB , and let O_1 and O_2 be the incenters of the triangles CAX and CBX . Prove that the circumcircle of the triangle XO_1O_2 intersects the circle Ω in a fixed point.

Proof. By $[F]$ we denote the area of figure F .

Let M be the midpoint of the arc BC . Applying trillium theorem we obtain that

$$MO_2 = MB = MC = MO,$$

where O is the incenter of the triangle ABC . In the same way A, O_1, O, B are on a circle with center N , the midpoint of the arc AC .



Let P lie on Ω such that $CP \parallel MN$ and T be the second intersection of OP with Ω (notice that T is constant). Then $MO = MC = NP$ and $MP = NC = NO$, therefore $MPNO$ is a parallelogram. It follows that $[MPT] = [NPT]$, whence $MP \cdot MT = NP \cdot NT$. Thus $NC \cdot MT = MC \cdot NT$, which shows that $NT : NO_1 = MT : MO_2$. But then $\triangle NO_1T \sim \triangle MO_2T$ (because it is true that $\sphericalangle O_1NT = \sphericalangle XNT = \sphericalangle XMT = \sphericalangle O_2MT$).

Moreover

$$\sphericalangle NTO_1 = \sphericalangle MTO_2,$$

and thus

$$\sphericalangle O_1XO_2 = \sphericalangle MTN = \sphericalangle O_1TO_2.$$

Therefore the quadrilateral XO_1O_2T is cyclic. This proves that the circumcircle of XO_1O_2 passes through the fixed point T or every position of X .

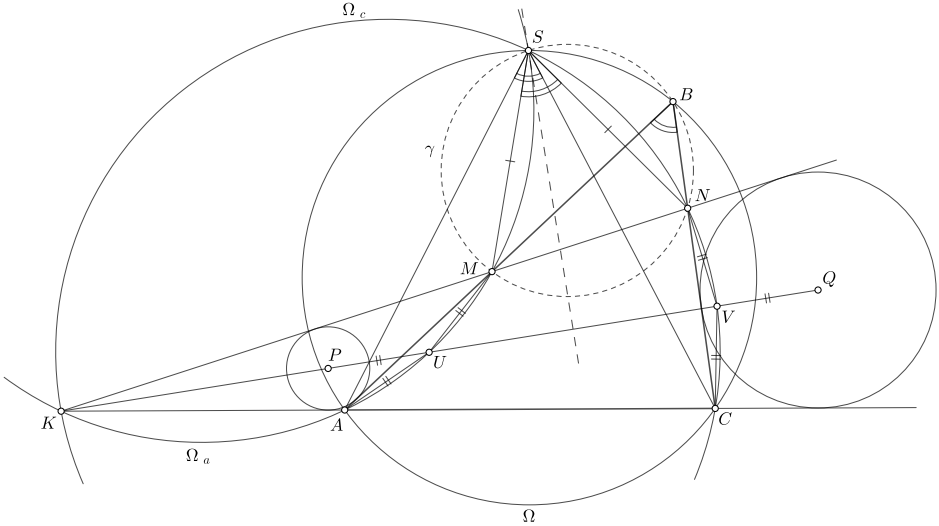
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Problem 12. Given is triangle ABC ($AB > BC$) inscribed in the circle Ω . On the sides AB and BC , points M and N are chosen, respectively, so that $AM = CN$. Lines MN and AC intersect at point K . Let P be the center of the inscribed circle of triangle AMK , and Q the center of the K -excircle of the triangle CNK . Prove that the midpoint of the arc AC , containing point B , of the circle Ω is equidistant from the points P and Q .

Proof. Recall the well-known lemma, which is general form of **trillium theorem**.

Lemma 1. Let L be the midpoint of arc YZ (not containing point X) of the circumscribed circle of the triangle XYZ . Let I be the center of the inscribed circle of the triangle XYZ , and I_x be the center of the X -excircle of this triangle. Then $LY = LZ = LI = LI_x$.

Let S be the midpoint of the arc ABC of the circle Ω . We have $SA = SC$ (since S lies on the perpendicular bisector of segment AC), $AM = CN$ and $\sphericalangle BCS = \sphericalangle BAS$ (angles inscribed, based on one arc). Thus triangles AMS and CNS are congruent. In particular they are congruent spirally, which means we can turn one with the center at the point S and the angle $\sphericalangle ASC = \sphericalangle ABC$ to get another. Hence we obtain $SM = SN$ and $\sphericalangle MSN = \sphericalangle ABC$. From the last equality of the angles it follows that the quadrilateral $MSBN$ is inscribed in some circle γ .



Denote by Ω_a and Ω_c the circles circumscribed on triangles AMS and CNS , respectively. They are also congruent spirally. Let U and V be the midpoints of the arcs AM and CN (not containing S) of these circles. From the congruency we get $SU = SV$ (because point S lies on the perpendicular bisector of UV) and $UA = VC$. Furthermore, from the angle chasing in circles Ω and γ , we have

$$\sphericalangle SAK = \sphericalangle SBC = \sphericalangle SMK,$$

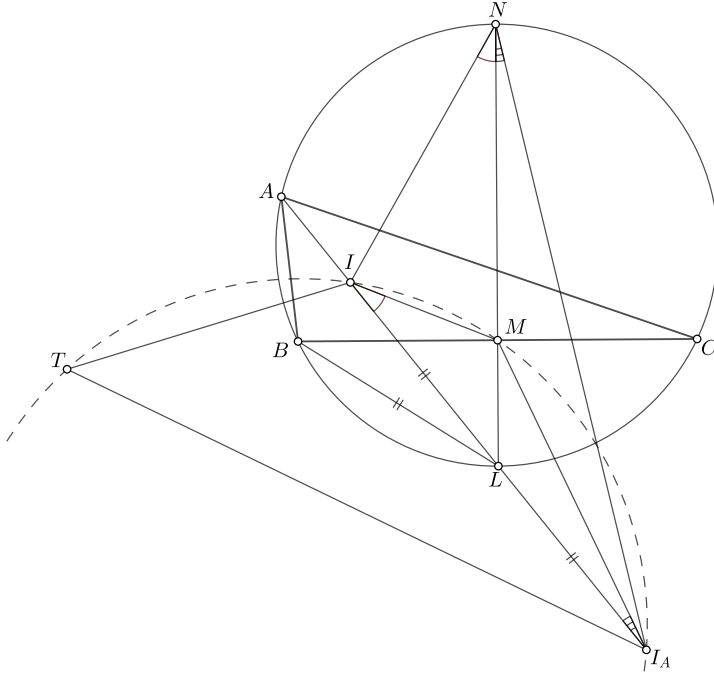
which means that K lies on Ω_a . Similarly, K lies on Ω_c .

It follows that the points U and V along with the points P and Q lie on the angle bisector of the angle CKN . By the trident lemma (for triangles KAM and KCN) we obtain that $UP = UA$ and $VQ = VC$. But since $UA = VC$, it means that the points P and Q are symmetric with respect to the perpendicular bisector to UV , on which the point S lies. So, S is equidistant from P and Q , as was desired.

Remark. Point S is *Miquel* point for the four lines BA , BC , AC and MN . $\square$

Problem 13. Denote by I the incenter of ABC , by I_A the A -excenter and by ω circumcircle of triangle ABC . Let M be the midpoint of segment CB and let N be the midpoint of arc BC , which contains point A . T is symmetric point to the point N in point reflection with respect to A . Prove that the quadrilateral I_AMIT is cyclic.

Proof. Let L be the midpoint of minor arc BC . It is well-known that $LI = LI_A = LB$. Notice that $LI^2 = LB^2 = LM \cdot LN$, because BM is an altitude in the right-angled triangle BLN . Thus triangles LMI and LNI are similar, which means that $\angle LIM = \angle LNI$. Analogously we can show that $\angle LI_A M = \angle LNI_A$.



Therefore $\angle IMI_A + \angle INI_A = 180^\circ$. As LN is a diameter of ω it follows that T and N are symmetric with respect to line II_A and therefore $\angle ITI_A = \angle INI_A$. This combined with the former equalities implies

$$\angle IMI_A + \angle ITI_A = 180^\circ,$$

which proves the required result. $\square$

Problem 14. The circle Γ has centre O , and BC is a diameter of Γ . Let A be a point which lies on Γ such that $\angle AOB < 120^\circ$. Let D be the midpoint of the arc AB which does not contain C . The line through O parallel to DA meets the line AC at I . The perpendicular bisector of OA meets Γ at E and at F . Prove that I is the incentre of the triangle CEF .

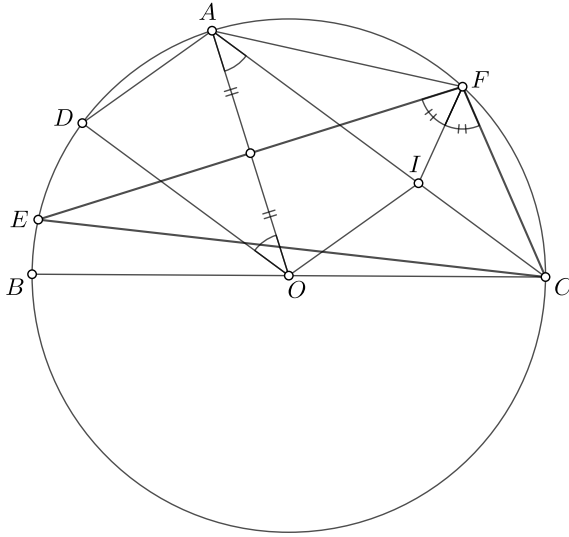
Proof. A is the midpoint of arc EAF , so CA bisects angle ECF . Now, since, $OA = OC$, $\angle AOD = \frac{1}{2}\angle AOB = \angle OAC$ so OD is parallel to IA and $ODAI$ is a parallelogram. Hence

$$AI = OD = OE = AF$$

since $OEAF$ (with diagonals bisecting each other at right angles) is a rhombus. Thus

$$\begin{aligned}\angle IFE &= \angle IFA - \angle EFA = \angle AIF - \angle ACE = \\ &= \angle AIF - \angle ICF = \angle IFC.\end{aligned}$$

Therefore, IF bisects angle EFC and I is the incentre of triangle CEF .



□

Problem 15. Let H be the orthocenter of the triangle ABC . Let M and N be the midpoints of the sides AB and AC , respectively. Assume that H lies inside the quadrilateral $BMNC$ and that the circumcircles of triangles BMH and CNH are tangent to each other. The line through H parallel to BC intersects the circumcircles of the triangles BMH and CNH in the points K and L , respectively. Let F be the intersection point of MK and NL and let J be the incenter of triangle MHN . Prove that $FJ = FA$.

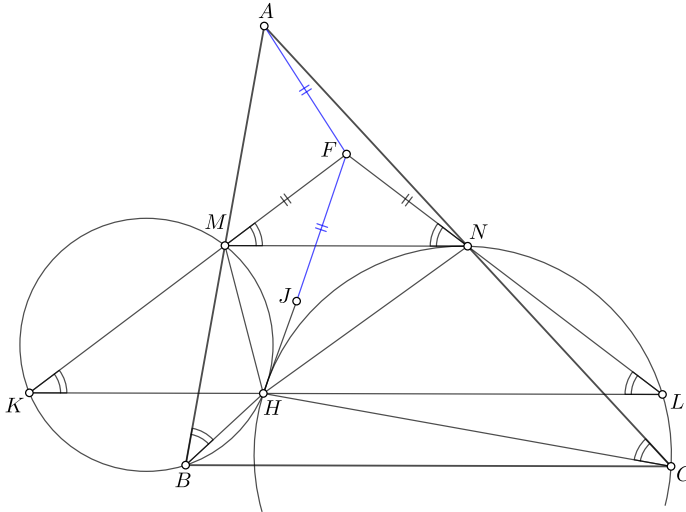
Proof. First we state some position considerations. Since there is an arc of the circumcircle of BHM outside the triangle ABC , it must happen that K and N lie on opposite sides of AM . Similarly, L and M lie on opposite sides of AN . Also, K and L lie on the same side of MN , and opposite to A . Therefore, F lies inside the triangle AMN .

Now, since H is the orthocenter of triangle ABC and the circumcircles of BMH and CNH are tangent we have

$$\begin{aligned} \angle ABH &= 90^\circ - \angle BAC = \angle ACH \implies \angle MHN = \angle MBH + \angle NCH = \\ &= 180^\circ - 2\angle BAC. \end{aligned}$$

So $\angle MBH = \angle MKH = \angle NCH = \angle NLH = 90^\circ - \angle BAC$ and, since $MN \parallel KL$, we have

$$\angle FMN = \angle FNM = 90^\circ - \angle BAC \implies \angle MFN = 2\angle BAC.$$



The above relations yield that the quadrilateral $MFNH$ is cyclic, with the vertices in this order around the circumference. Since $FM = FN$, $\angle MFN = 2\angle BAC$ and F is the correct side of MN we have that the point F is the circumcenter of triangle AMN , and thus $FA = FM = FN$.

Since the quadrilateral $MFNH$ is cyclic, $FM = FN$ and H lies on the correct side of MN , we have that H , J and F are collinear. According to trillium theorem, $FJ = FM = FN$. So $FJ = FA$. $\square$

Problem 16. A trapezoid $ABCD$ in which $AB \parallel CD$ and $AB > CD$, is circumscribed. The incircle of the triangle ABC touches the lines AB and AC at the points M and N , respectively. Prove that the incenter of the trapezoid $ABCD$ lies on the line MN .

Proof. Let I be the incenter of triangle ABC and R be the common point of the lines BI and MN . Since

$$\angle ANM = 90^\circ - \frac{1}{2}\angle MAN$$

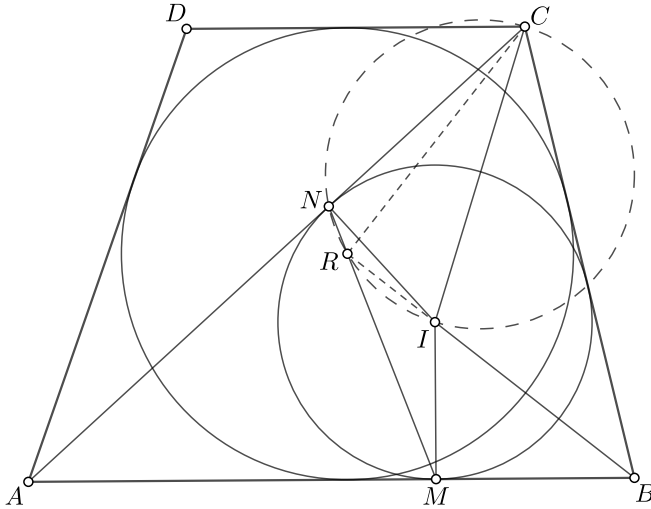
and

$$\angle BIC = 90^\circ + \frac{1}{2}\angle MAN$$

the quadrilateral $IRNC$ is cyclic. It follows that $\angle INC = \angle BRC = 90^\circ$ and therefore

$$\angle BCR = 90^\circ - \angle CBR = 90^\circ - \frac{1}{2}(180^\circ - \angle BCD) = \frac{1}{2}\angle BCD.$$

Hence CR is the angle bisector of angle DCB and R is the incenter of the trapezoid.



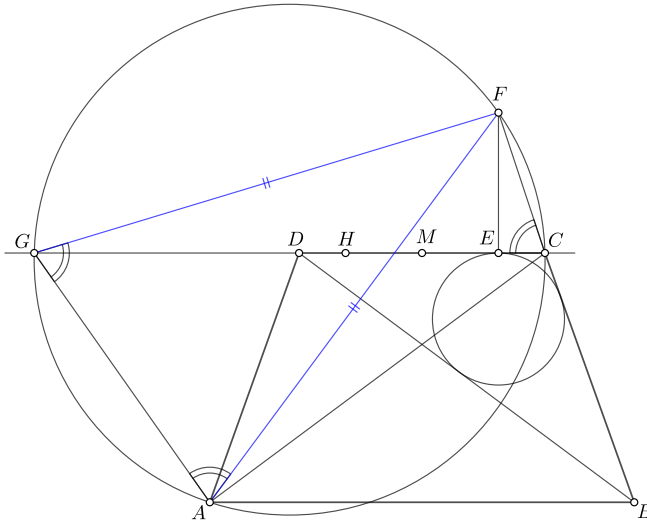
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Problem 17. Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. The inscribed circle ω of triangle BCD meets CD at E . Let F be a point on the (internal) angle bisector of angle DAC such that $EF \perp CD$. Let the circumscribed circle of triangle ACF meet line CD at C and G . Prove that the triangle AFG is isosceles.

Proof. Let the incircle of triangle ADC touch DC at H , and let F' be the A -excenter of triangle ADC . Since $\triangle ADC \equiv \triangle BCD$, E and H are reflections with respect to the midpoint M of segment DC . Hence E is the point of contact of the A -excircle of $\triangle ADC$ with DC . It means that $EF' \perp CD$. Since both F and F' lie on angle bisector of angle DAC , we obtain that $F' = F$. Thus, we have

$$\angle GAF = \angle GCF = 90^\circ - \frac{1}{2}\angle ACD = 180^\circ - \angle ACF = \angle AGF,$$

so $\triangle AFG$ is isosceles, as desired.



□

Problem 18. Let N be the midpoint of arc AC containing point B of the circumcircle of triangle ABC , let M be the midpoint of segment AC and let I_1, I_2 be the incentres of triangles ABM and CBM . Prove that quadrilateral points $I_1 I_2 B N$ is cyclic.

Proof. Let I_3 be the reflection of I_2 respect to the line MN . We will show that I_3 is the **isogonal conjugate** of I_1 in triangle AMN . Notice that it is sufficient to prove that $\sphericalangle I_3 M N = \sphericalangle I_1 M A$ and $\sphericalangle I_3 A M = \sphericalangle I_1 A N$. Note that

$$\sphericalangle I_3 M N = \sphericalangle I_2 M N = 90^\circ - \sphericalangle I_1 M N = \sphericalangle I_1 M A,$$

and

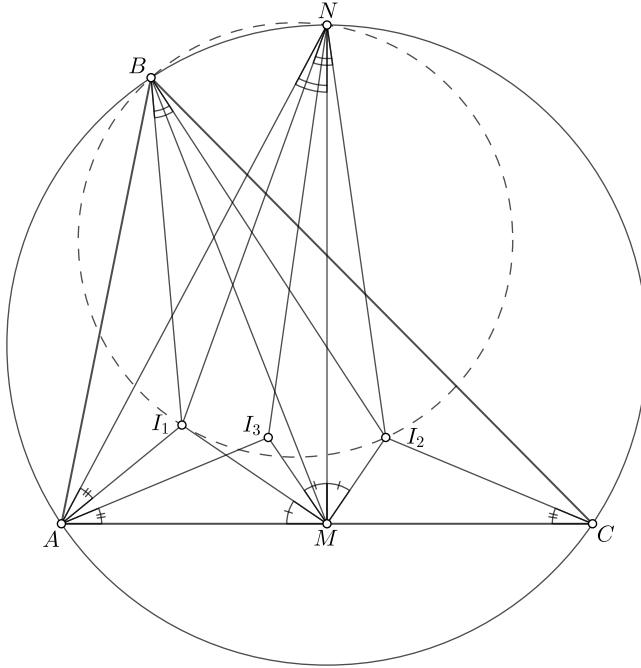
$$\sphericalangle I_3 A M = \sphericalangle I_2 C M = \frac{1}{2} \sphericalangle B C A = \sphericalangle I_1 A N.$$

Hence we are done with proof that I_3 is the **isogonal conjugate** of I_1 in triangle AMN .

Now, using the above conclusion, one can easily calculate:

$$\begin{aligned} \sphericalangle I_2 N I_1 &= \sphericalangle I_2 N M + \sphericalangle M N I_1 = \sphericalangle I_3 N M + \sphericalangle M N I_1 = \sphericalangle I_1 N A + \sphericalangle M N I_1 = \\ &= \sphericalangle M N A = \frac{1}{2} \sphericalangle A B C = \sphericalangle I_2 B I_1 \end{aligned}$$

So indeed points I_1, I_2, B, N are concyclic.



□

Problem 19. Let $ABCD$ be a convex quadrilateral with $BA \neq BC$. Denote the incircles of triangles ABC and ADC by ω_1 and ω_2 respectively. Suppose that there exists a circle ω tangent to ray BA beyond A and to the ray BC beyond C , which is also tangent to the lines AD and CD . Prove that the common external tangents to ω_1 and ω_2 intersect on ω .

Proof. Let M, N, P, Q be the points of tangency of ω with AB, BC, CD , and DA , respectively. We have that

$$\begin{aligned} AB + AD &= AB + AQ - QD = AB + AM - DP = BM - CP + CD = \\ &= BN - CN + CD = BC + CD. \end{aligned}$$

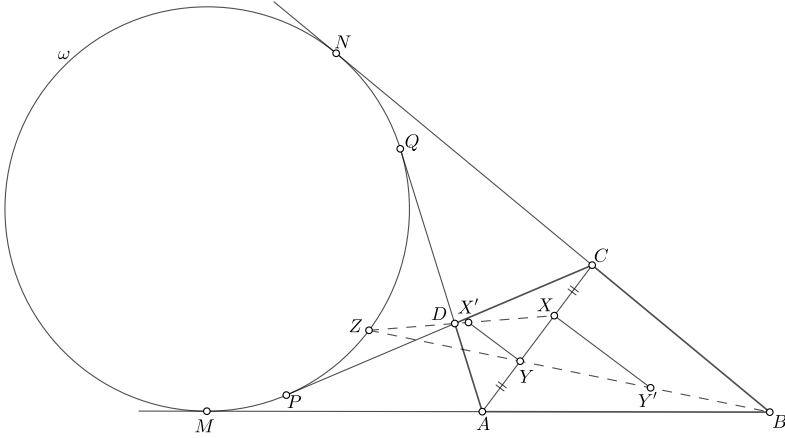
Denote by X and Y the points of tangency of ω_1 and ω_2 with AC . Then we have

$$AB = AX + BC - CX$$

and

$$AD = AY + CD - CY.$$

Together with $AB + AD = BC + CD$ this yields to $AX - CX = CY - AY$. Since $AX + CX = CY + AY$ we conclude that $AX = CY$ hence Y is the point of tangency of AC and the excircle ω_B of triangle ABC that corresponds to B . Similarly, the excircle ω_D corresponding to D of triangle ADC passes through X .



Consider the homothety that maps ω_B to ω . Denote by Z the image of Y under this homothety. Z belongs to the tangent of ω that is parallel to AC . Therefore Z is the image of X under the homothety with center D that maps ω_D to ω . Denote by X' and Y' the intersections of DX and BY with ω_2 and ω_1 respectively. Circles ω_1 and ω_B are homothetic with center B , hence Y' is the image of Y under this homothety. Moreover, Y' belongs to the tangent of ω_1 that is parallel to AC . This implies that XY' is a diameter of ω_1 . Similarly, $X'Y$ is a diameter of ω_2 . This implies that $X'Y \parallel XY'$ which means that triangles $ZX'Y$ and ZXY' are similar and Z is the center of homothety that maps ω_2 to ω_1 . This finishes the proof of the required statement. $\square$

Problem 20. Let I be the incentre of a non-equilateral triangle ABC , I_A be the A -excentre, I'_A be the reflection of I_A in BC , and l_A be the reflection of line AI'_A in AI . Define points I_B , I'_B and line l_B analogously. Let P be the intersection point of l_A and l_B .

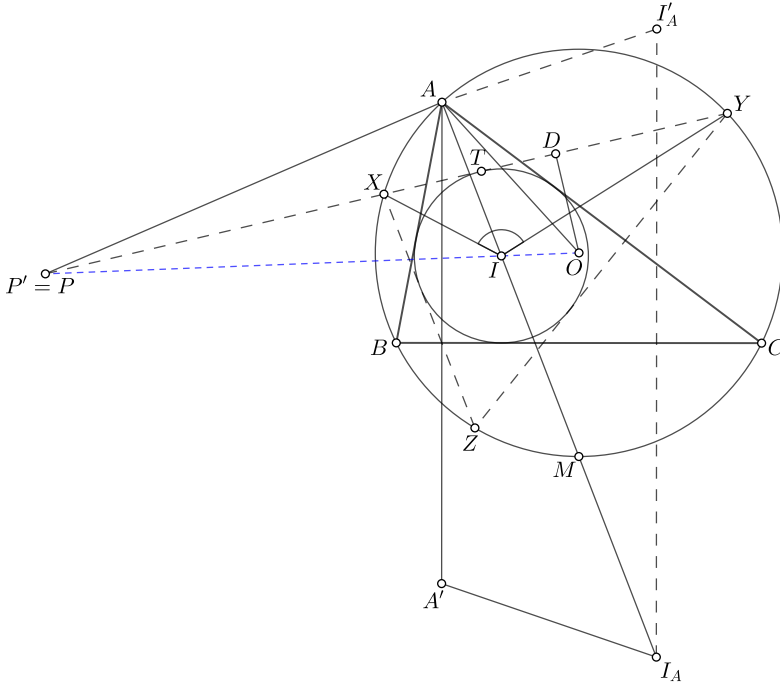
(a) Prove that P lies on line OI where O is the circumcentre of triangle ABC .

(b) Let one of the tangents from P to the incircle of triangle ABC meet the circumcircle at points X and Y . Show that $\sphericalangle XIY = 120^\circ$.

Proof. (a) Let A' be the reflection of A in BC and let M be the second intersection of line AI and the circumcircle Γ of triangle ABC . As triangles ABA' and AOC are isosceles with $\sphericalangle ABA' = 2\sphericalangle ABC = \sphericalangle AOC$, they are similar to each other. Also, triangles ABI_A and AIC are similar. Therefore we have

$$\frac{AA'}{AI_A} = \frac{AA'}{AB} \cdot \frac{AB}{AI_A} = \frac{AC}{AO} \cdot \frac{AI}{AC} = \frac{AI}{AO}.$$

Together with $\sphericalangle A'AI_A = \sphericalangle IAO$ (we are using the fact that O and H are isogonal conjugated), we find that triangles $AA'I_A$ and AIO are similar.



Denote by P' the intersection of line AP and line OI . Using directed angles, we obtain:

$$\begin{aligned} \sphericalangle MAP &= \sphericalangle I'_A AI_A = \sphericalangle I'_A AI'_A - \sphericalangle I_A AA' = \sphericalangle AA' I_A - \sphericalangle (AM, OM) = \\ &= \sphericalangle AIO - \sphericalangle AMO = \sphericalangle MOP'. \end{aligned}$$

This shows that points M , O , A , P' are concyclic.

Denote by R and r the circumradius and inradius of triangle ABC . Then

$$IP' = \frac{IA \cdot IM}{IO} = \frac{IO^2 - R^2}{IO}$$

is independent of A . Hence, BP also meets line OI at the same point P' so that $P' = P$, and P lies on OI .

(b) By Poncelets Porism, the other tangents to the incircle of triangle ABC from X and Y meet at a point Z on Γ . Let T be the touching point of the incircle to XY , and let D be the midpoint of XY . We have

$$\begin{aligned} OD &= IT \cdot \frac{OP}{IP} = r \left(1 + \frac{OI}{IP} \right) = r \left(1 + \frac{OI^2}{OI \cdot IP} \right) = r \left(1 + \frac{R^2 - 2Rr}{R^2 - IO^2} \right) = \\ &= r \left(1 + \frac{R^2 - 2Rr}{2Rr} \right) = \frac{R}{2} = \frac{OX}{2}. \end{aligned}$$

This shows $\angle XZY = 60^\circ$ and hence $\angle XIY = 120^\circ$. □

2.2. Power of a point.

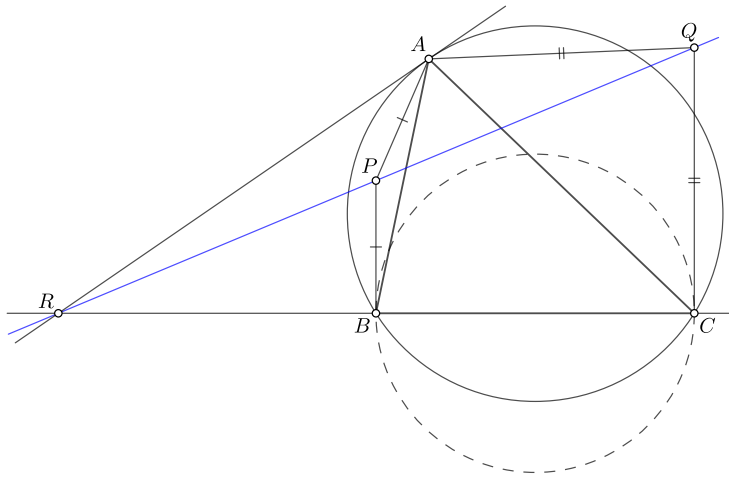
Problem 1. *Given is triangle ABC . Points P and Q were chosen such that $\sphericalangle PBC = \sphericalangle QCB = 90^\circ$ and $AP = PB$, $AQ = CQ$.*

Tangent to circumcircle of ABC passing through point A intersects BC at R . Prove that P, Q and R are collinear.

Proof. Consider radical axis of point A , circumcircle ω of ABC and circle with diameter BC . Then radical axis of

- A and ω is line AA ;
- A and circle with diameter BC is line PQ ;
- ω and circle with diameter BC is line BC .

Therefore lines AA , PQ and BC intersect at one point - namely R , thus P, Q, R are collinear.



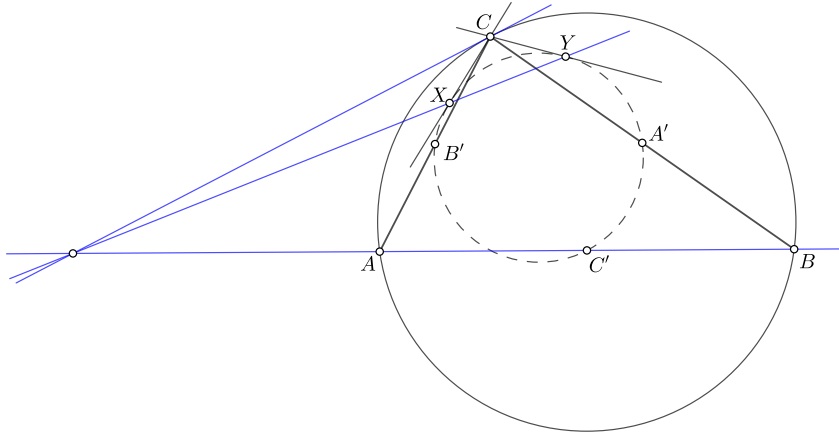
□

Problem 2. Let CX , CY be the tangents from vertex C of triangle ABC to the circle passing through the midpoints of its sides. Prove that lines XY , AB and the tangent at point C to the circumcircle ω of ABC concur at one point.

Proof. Consider the homothety with center C and factor $\frac{1}{2}$. After this homothety:

- (1) line XY is transformed into the radical axis $X'Y'$ of point C and the circle passing through the midpoints A' , B' , C' of BC , CA , AB ;
- (2) line $A'B'$ is transformed into the radical axis of ω and circumcircle of $A'B'C'$,
- (3) line CC is the radical axis of ω and point C .

Thus lines $X'Y'$, $A'B'$ and CC concur. Using the inverse homothety we obtain the assertion of the problem.



□

Problem 3. An acute triangle ABC in which $AB < AC$ is given. Points E and F are feet of its heights from B and C , respectively. The line tangent in point A to the circle circumscribed on ABC crosses BC at P . The line parallel to BC that goes through point A crosses EF at Q . Prove PQ is perpendicular to the median from A of triangle ABC .

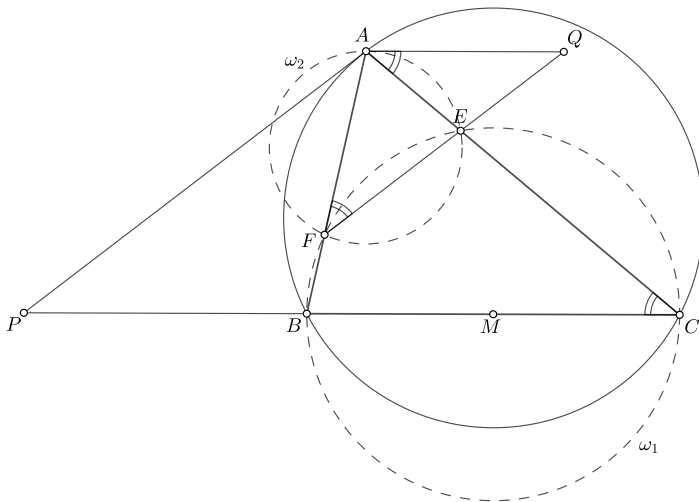
Proof. Notice that quadrilateral $EFBC$ is cyclic, because $\sphericalangle BEC = \sphericalangle BFC = 90^\circ$. Denote circumcircle of $EFBC$ by ω_1 . Moreover

$$\sphericalangle EFA = 180^\circ - \sphericalangle BFE = \sphericalangle ACB = \sphericalangle EAQ,$$

Last equality comes from fact that $AQ \parallel BC$.

It's not hard to notice that circumcircle ω_2 of triangle AFE is tangent to line AQ . It follows from radical axis concurrence theorem applied to point A and circles ω_1 and ω_2 that point Q lies on radical axis of point A and ω_1 .

Once again applying this theorem to point A , circumcircle of ABC and ω_1 we obtain that P lies on radical axis of A and ω_1 .



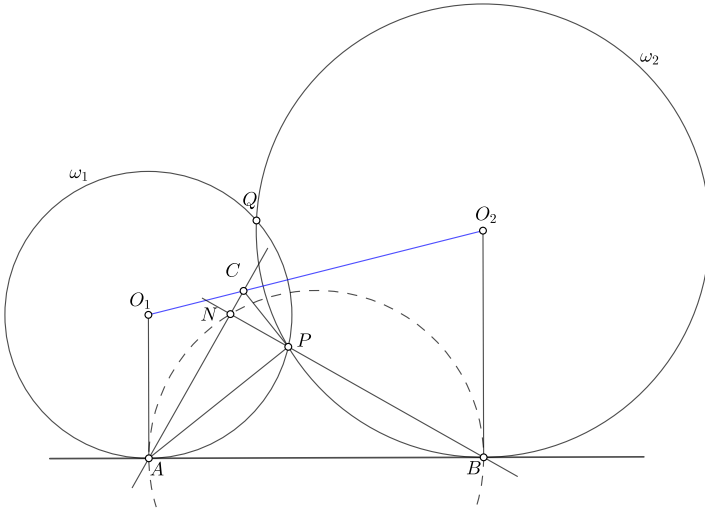
Thus line PQ is radical axis of point A and circle ω_1 . In particular PQ is perpendicular to line connecting centers of this circles - namely A and M . Hence $PQ \perp AM$ and we are done. $\square$

Problem 4. Let circles ω_1 and ω_2 , with centres in O_1, O_2 , respectively, intersect at two distinct points P and Q . Their common tangent, closer to point P , touches the circles at A, B respectively. Let the perpendicular from A to the line BP meet O_1O_2 at C . Prove that $\angle APC = 90^\circ$.

Proof. Let $N = AC \cap BP$. Powers of point O_1 with respect to point P and circumcircle of ANB are equal (notice that O_1A is tangent to circumcircle of ANB). Similarly it holds for O_2 . Thus we can obtain that O_1O_2 is the radical axis of point P and circumcircle of ANB . Since C lies on O_1O_2 we get

$$CN \cdot CA = CP^2.$$

From now on it is easy to get that $\angle APC = 90^\circ$.



□

Problem 5. The circumradius and the inradius of triangle ABC are equal to R and r . O, I are the centers of respective circles. External bisector of angle ACB intersect AB at point P . Point Q is the projection of P to line OI . Find the length of segment OQ .

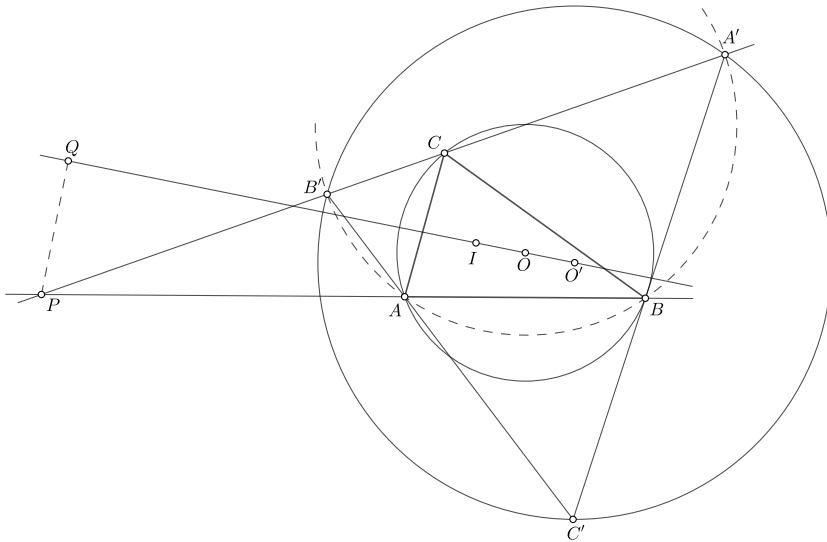
Proof. Denote by A', B', C' the A, B, C -excenters of triangle ABC , respectively. Then it is well known that I is the orthocenter of triangle $A'B'C'$ and A, B, C are the bases of its altitudes and so the circumcircle of ABC is the Euler circle of $A'B'C'$. Thus the circumradius of $A'B'C'$ is $2R$, and its circumcenter O' is the reflection of I in O .

Furthermore points A, B, A', B' lie on the circle. Line AB is the common chord of this circle and the circumcircle of ABC , and the external bisector of C is the common chord of this circle and the circumcircle of $A'B'C'$. So P is the radical center of three circles, and line PQ is the radical axis of circumcircles of ABC and $A'B'C'$. Therefore from power of point:

$$OQ^2 - R^2 = (OQ + OO')^2 - 4R^2.$$

From **Euler's theorem** $OO' = OI = \sqrt{R^2 - 2Rr}$, thus we obtain

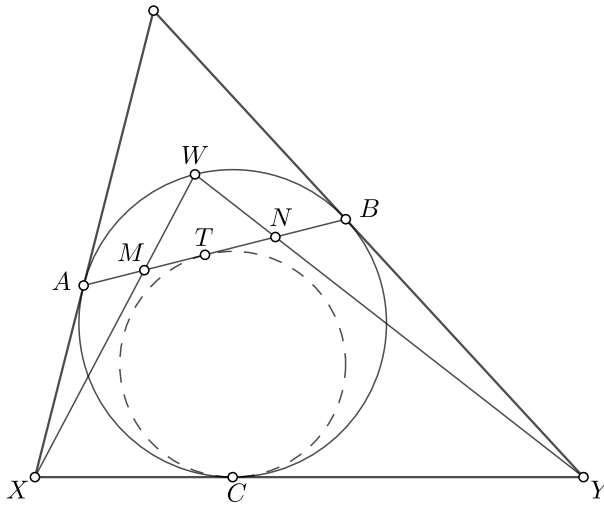
$$OQ = \frac{R(R+r)}{\sqrt{R^2 - 2Rr}}.$$



□

Problem 6. A circle Ω , its chord AB and the midpoint W of the minor arc AB are given. Take an arbitrary point C on the major arc AB . The tangent to the circle at C meets the tangents at A and B at points X and Y , respectively. Lines WX and WY meet AB at points N and M respectively. Prove that the length of segment NM does not depend on point C .

Proof. Consider circle ω , touching XY at C and touching AB (at point T). It is easy to see that WX is the radical axis of point A and ω (one just need to notice that WT and C are collinear by homothety and apply the definition of homothety), i.e. it passes through the midpoint N of segment AT . Similarly WY passes through the midpoint M of segment TB . Thus $MN = \frac{AB}{2}$.

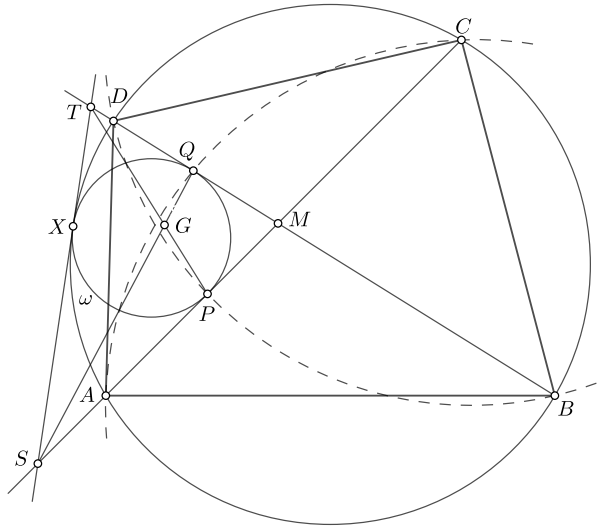


□

Problem 7. The diagonals of a cyclic quadrilateral $ABCD$ meet at point M . A circle ω touches segments MA and MD at points P , Q respectively and is tangent to the circumcircle of $ABCD$ at point X . Prove that X lies on the radical axis of circumcircles of ACQ and BDP .

Proof. Let l be the tangent at X to the circumcircle of $ABCD$; and let l meet AC and BD at the points S and T , respectively.

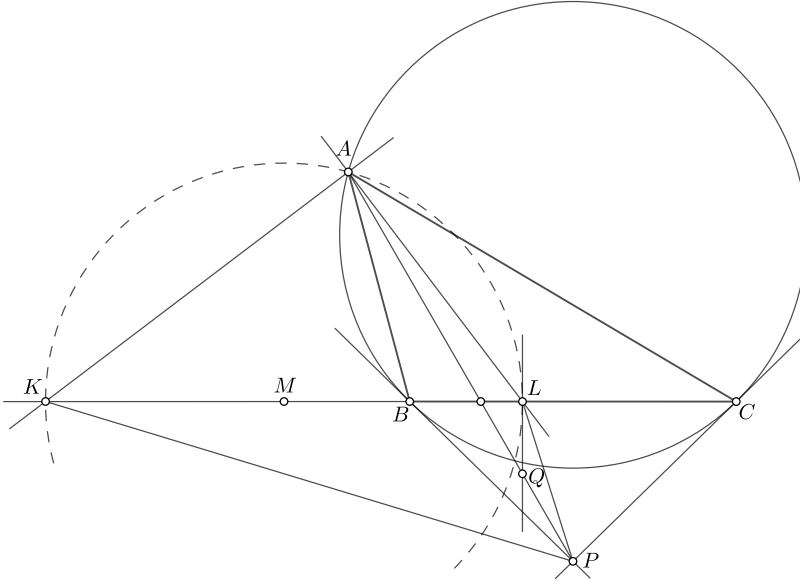
Then SM is the radical axis of circumscribed circles of $ABCD$ and ACQ , ST is the radical axis of circumscribed circles $ABCD$ and ω . It means that S is the radical center of circumcircles of $ABCD$, ACQ and ω , hence SQ is the radical axis of circles ACQ and ω (because Q lies on the intersection of circles ACQ and ω !). Similarly TP is the radical axis of circumscribed circles of BDP and ω .



Therefore the common point G of SQ and TP is the radical center of circles ACQ , BDP and ω . On the other hand M is the radical center of circumcircles of ACQ , BDP and $ABCD$. Thus MG is the radical center of circles ACQ and BDP . But MG passes also through X , because G is the **Gergonne's point** of triangle MST . $\square$

Problem 8. Let L and K be the feet of the internal and the external bisectors of angle BAC in a triangle ABC , respectively. Let P be the common point of the tangents to the circumcircle of the triangle at B and C . The perpendicular from L to BC meets AP at point Q . Prove that Q lies on the medial line of triangle LKP .

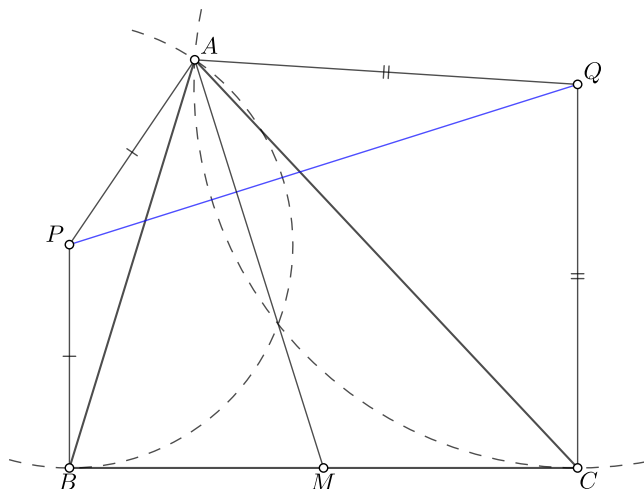
Proof. Since BC is the polar of P with respect to the circumcircle ω of triangle ABC we obtain that P lies on the polar of L . Because $(K, L; B, C) = -1$, then K also lies on the polar of L . Therefore KP is the polar of L with respect to ω . The medial line of triangle KLP is of course the radical axis of ω and point L . We will prove that Q also lies on this axis.



Let M be the midpoint of segment KL . Since M is the center of circle AKL which is perpendicular to ω . M lies on the polar of A . But M also lies on the polar of P , thus AP both is the polar of M with respect to ω and the common chord of ω and circle AKL . But LQ is the radical axis of circle AKL and L , therefore, Q is the common point of three radical axes and hence lies on medial line of triangle KLP . $\square$

Problem 9. Lines b and c passing through vertices B and C of triangle ABC are perpendicular to sideline BC . The perpendicular bisectors of AC and AB meet b and c at points P and Q , respectively. Prove that the line PQ is perpendicular to median AM of triangle ABC .

Proof. Construct a circle centred at P and passing through A . It is tangent to BC at B , because $PA = PB$. Similarly, the circle centred at Q and passing through A is tangent to BC at C . The powers of M with respect to these circles are equal, so the radical axis of these circles is AM . It is perpendicular to the line connecting centres of these circles, namely line PQ . Hence we are done.



□

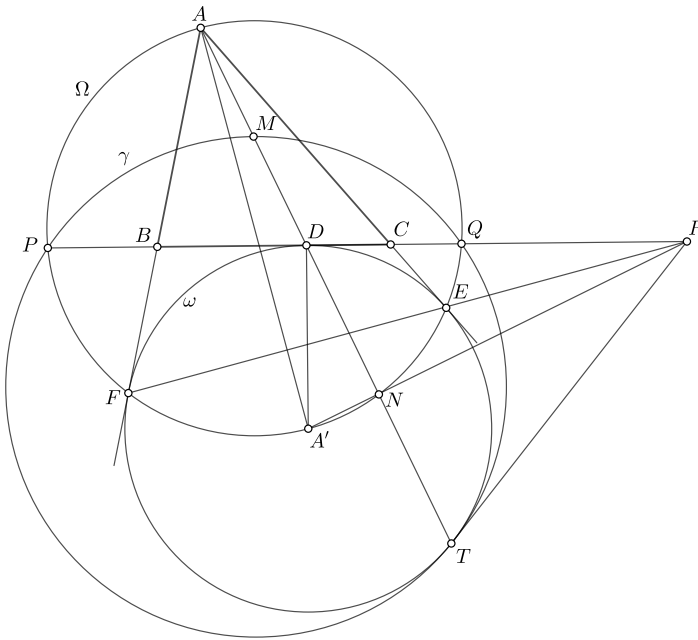
Problem 10. In triangle ABC , let ω be the excircle opposite to A . Let D, E and F be the points where ω is tangent to BC, CA , and AB , respectively. The circle AEF intersects line BC at P and Q . Let M be the midpoint of AD . Prove that the circle MPQ is tangent to ω .

Proof. Denote by Ω the circle $AEFPQ$, and denote by γ the circle PQM . Let the line AD meet ω again at $T \neq D$. We will show that γ is tangent to ω at T .

We first prove that points P, Q, M, T are concyclic. Let A' be the center of ω . Since $A'E \perp AE$ and $A'F \perp AF$, AA' is a diameter in Ω . Let N be the midpoint of DT ; from $A'D = A'T$ we can see that $\angle A'NA = 90^\circ$ and therefore N also lies on the circle Ω . Now, from the power of D with respect to the circles γ and Ω we get

$$DP \cdot DQ = DA \cdot DN = 2DM \cdot \frac{DT}{2} = DM \cdot DT,$$

so P, Q, M, T are concyclic.



If $EF \parallel BC$, then ABC is *isosceles* and the problem is now immediate by symmetry. Otherwise, let the tangent line to ω at T meet line BC at point R . The tangent line segments RD and RT have the same length, so $A'R$ is the perpendicular bisector of DT ; since $ND = NT$, N lies on this perpendicular bisector.

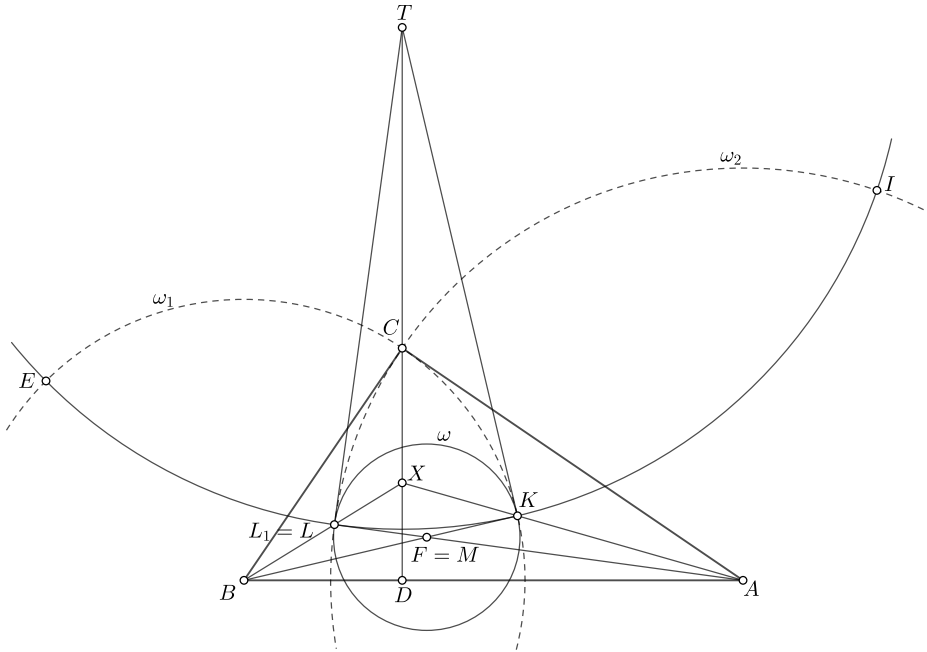
In right triangle $A'RD$, $RD^2 = RN \cdot RA' = RP \cdot RQ$, in which the last equality was obtained from the power of R with respect to Ω . Hence $RT^2 = RP \cdot RQ$, which implies that RT is also tangent to γ . Because RT is a common tangent to ω and γ , these two circles are tangent at T . $\square$

Problem 11. Let ABC be a triangle with $\angle BCA = 90^\circ$, and let D be the foot of the altitude from C . Let X be a point in the interior of the segment CD . Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. Let M be the point of intersection of AL and BK . Show that $MK = ML$.

Proof. Consider the circles $\omega_1(B, BC)$, $\omega_2(A, AC)$ and $\omega(F, FK)$, where the circle ω has its center on BK and it is internally tangential to the other two circles ω_1, ω_2 at K and L_1 , respectively. The radical axes of the three circles will be intersected at point T of altitude CD . We have that $X = CD \cap AK$.

Let the circle (T, TK) , where $TK = TL_1$, intersects ω_2 at point I . The point B belongs to the radical axis L_1I , since the triangle is a right one at the vertex C and $BC = BK$.

Similarly, the straight line AK is the radical axis of the circles $\omega_1, (T, TK)$. Thus, because of uniqueness of the points, we deduce that $L_1 = L$ and hence $F = M$. This ends the proof.



□

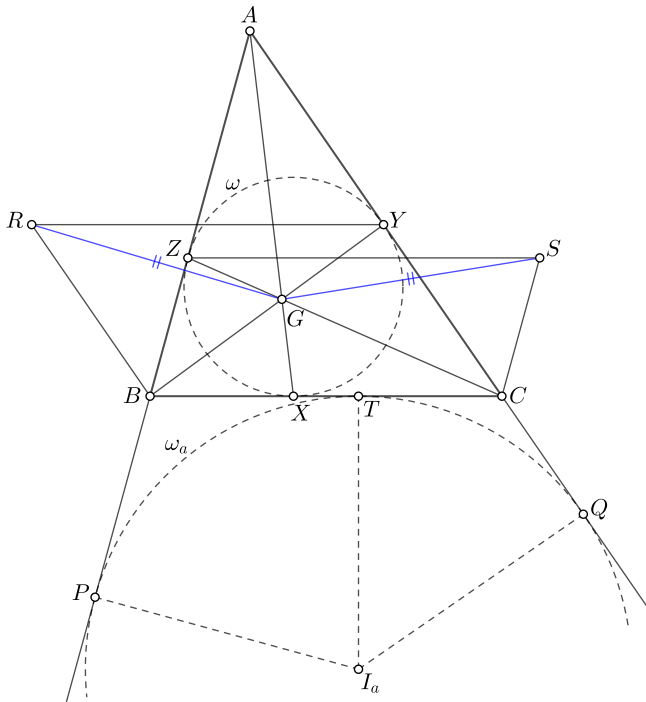
Problem 12. Let ABC be a triangle. The incircle of ABC touches the sides AB and AC at the points Z and Y , respectively. Let G be the point where the lines BY and CZ meet, and let R and S be points such that the two quadrilaterals $BCYR$ and $BCSZ$ are parallelogram. Prove that $GR = GS$.

Proof. Denote by ω the incircle and by ω_a the A -excircle of triangle ABC . Let ω and ω_a touch the side BC at the points X and T , respectively; let ω_a touch the lines AB and AC at the points P and Q , respectively. We use several times the fact that opposing sides of a parallelogram are of equal length, that points of contact of the excircle and incircle to a side of a triangle lie symmetric with respect to the midpoint of this side and that segments on two tangents to a circle defined by the points of contact and their point of intersection have the same length. So we conclude

$$ZP = ZB + BP = XB + BT = BX + CX = ZS$$

and

$$CQ = CT = BX = BZ = CS.$$



So for each of the points Z, C , their distances to S equal the length of a tangent segment from this point to ω_a . It is well-known, that all points with this property lie on the line ZC , which is the radical axis of S and ω_a . Similar arguments yield that BY is the radical axis of R and ω_a . So the point of intersection of ZC and BY , which is G by definition, is the radical center of R, S and ω_a , from which the claim $GR = GS$ follows immediately. $\square$

Problem 13. Let $ABCD$ be a convex quadrilateral whose sides AD and BC are not parallel. Suppose that the circles with diameters AB and CD meet at points E and F inside the quadrilateral. Let ω_E be the circle through the feet of the perpendiculars from E to the lines AB, BC and CD . Let ω_F be the circle through the feet of the perpendiculars from F to the lines CD, DA and AB . Prove that the midpoint of the segment EF lies on the line through the two intersections of ω_E and ω_F .

Proof. Denote by P, Q, R , and S the projections of E on the lines DA, AB, BC , and CD respectively. The points P and Q lie on the circle with diameter AE , so $\angle QPE = \angle QAE$; analogously, $\angle QRE = \angle QBE$. So $\angle QPE + \angle QRE = \angle QAE + \angle QBE = 90^\circ$. By similar reasons, we have $\angle SPE + \angle SRE = 90^\circ$, hence we get

$$\angle QPS + \angle QRS = 90^\circ + 90^\circ = 180^\circ,$$

and the quadrilateral $PQRS$ is inscribed in ω_E . Analogously, all four projections of F onto the sides of $ABCD$ lie on ω_F .

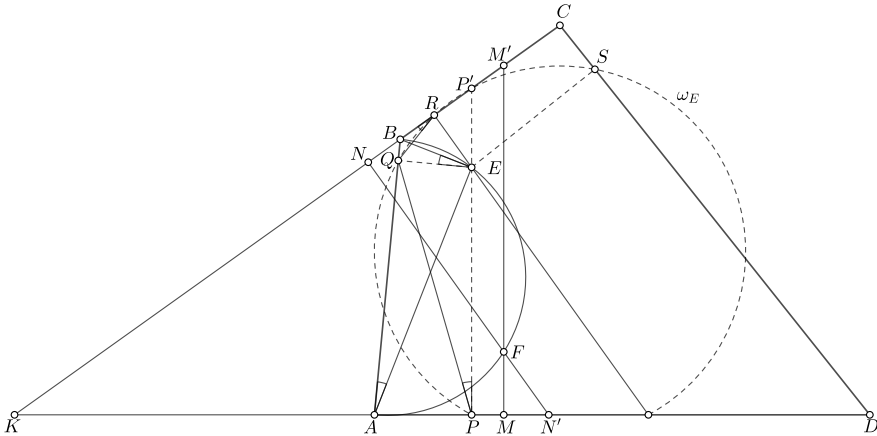
Denote by K the meeting point of the lines AD and BC . Due to the arguments above, there is no loss of generality in assuming that A lies on segment DK . Suppose that $\angle CKD \geq 90^\circ$; then the circle with diameter CD covers the whole quadrilateral $ABCD$, so the points E, F cannot lie inside this quadrilateral. Hence our assumption is wrong. Therefore, the lines EP and BC intersect at some point P' , while the lines ER and AD intersect at some point R' .

We claim that the points P' and R' also belong to ω_E . Since the points R, E, Q, B are concyclic,

$$\angle QRK = \angle QEB = 90^\circ - \angle QBE = \angle QAE = \angle QPE.$$

So $\angle QRK = \angle QPP'$, which means that the point P' lies on ω_E . Analogously, R' also lies on ω_E .

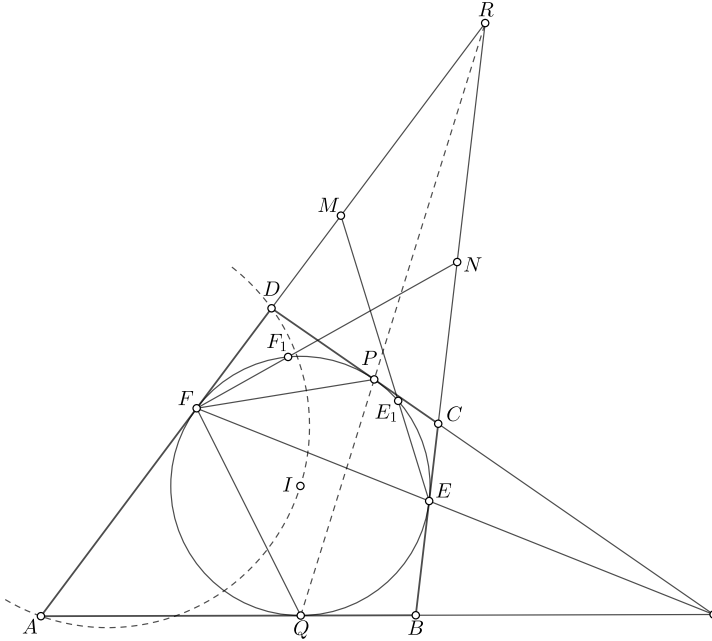
In the same manner, denote by M and N the projections of F on the lines AD and BC respectively, and let $M' = FM \cap BC$, $N' = FN \cap AD$. By the same arguments, we obtain that the points M' and N' belong to ω_F .



Problem 14. Let $ABCD$ be a circumscribed quadrilateral. Its incircle ω touches sides BC and DA at points E and F respectively. It is known that lines AB , FE and CD concur. The circumcircles of triangles AED and BFC meet ω for the second time at points E_1 and F_1 . Prove that $EF \parallel E_1F_1$.

Proof. Let R be the common point of BC and AD . Then R and the touching points P and Q of the incircle with two remaining sides are collinear.

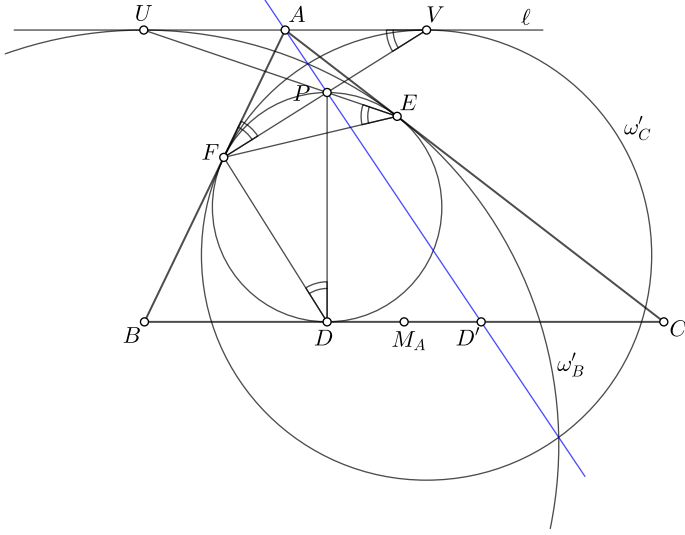
Let EE_1 meet AD at point M . Consider three circles: the incircle of $ABCD$, AED and AID , where I is the incenter of $ABCD$. It is clear that the radical axis of AID and the incircle is the medial line of triangle FPQ . Since two remaining radical axes meet at M we obtain that $RM = MF$.



Similarly, FF_1 meets BC at point N , such that $RN = NE$. Therefore lines EE_1 and FF_1 are symmetric with respect to the bisector of angle ERF . Thus points E_1 and F_1 are also symmetric and EFF_1E_1 is an isosceles trapezoid. $\square$

Problem 15. Circles ω_B and ω_C are excircles of a triangle ABC . The circle ω'_B is symmetric to ω_B with respect to the midpoint of AC , the circle ω'_C is symmetric to ω_C with respect to the midpoint of AB . Prove that the radical axis of ω'_B and ω'_C halves the perimeter of ABC .

Proof. Let the incircle of triangle ABC be tangent to lines BC , CA , and AB , at D , E , and F , respectively. Denote by D' the reflection of D with respect to the midpoint of BC . It is well-known that AD' passes through the antipodic point P of D with respect to the incircle of triangle ABC .



Let ℓ be the line through A parallel to BC . Notice ω'_C is the circle tangent to AB at F and tangent to ℓ ; similarly ω'_B is the circle tangent to AC at E and tangent to ℓ . Denote by U and V the intersections of EP and FP with ℓ , respectively. Note that

$$\sphericalangle FAV = \sphericalangle FAC + \sphericalangle CAV = \sphericalangle A + \sphericalangle C = 180^\circ - \sphericalangle B,$$

while

$$\sphericalangle VFA = \sphericalangle PFA = \sphericalangle FDP = \frac{1}{2} \sphericalangle B,$$

so $AF = AV$. Analogously $AE = AU$. Hence, U and V must be the tangency points of ω'_B and ω'_C with ℓ , respectively. Furthermore,

$$\sphericalangle UVF = \sphericalangle AVF = \sphericalangle VFA = \sphericalangle PEF = \sphericalangle UEF,$$

so $UVEF$ is cyclic. Thus we conclude that $PU \cdot PE = PV \cdot PF$. Therefore P lies on the radical axis of ω'_B and ω'_C . Since $AF = AE$, A also lies on this radical axis. Thence AP is the radical axis of ω'_B and ω'_C . Recall that D' lies on line AP . Thus

$$AB + BD' = AB + \frac{CA + CB - AB}{2} = \frac{AB + BC + CA}{2}.$$

Hence line AD' , namely the radical axis of ω'_B and ω'_C , halves the perimeter of ABC and we are done. $\square$

Problem 16. *Let ABC be a scalene triangle with circumcircle Γ . Let M be the midpoint of BC . A variable point P is selected in the line segment AM . The circumcircles of triangles BPM and CPM intersect Γ again at points D and E , respectively. The lines DP and EP intersect (a second time) the circumcircles to triangles CPM and BPM at X and Y , respectively. Prove that as P varies, the circumcircle of triangle AXY passes through a fixed point T distinct from A .*

2.3. SSS - Similarity, (Spiral) Similarity & Symmedians.

Problem 1. Let ABC be an acute triangle in which $AB \neq AC$. Point J is an A -excentre of triangle ABC ; A -excircle touches segment BC at E . Lines BC and AJ intersect at D . Circumcircles of triangles ABC and ADE intersect at $F \neq A$. Prove that $\angle AFJ = 90^\circ$.

Proof. Without loss of generality assume that $AB > AC$; then points F and C lie on the opposite side of line AB . Let A -excircle touch lines AB , AC at points P and Q , respectively. Since

$$\angle EFA = \angle ADC = \angle CBA + \angle BAD = \angle CFA + \angle BAD,$$

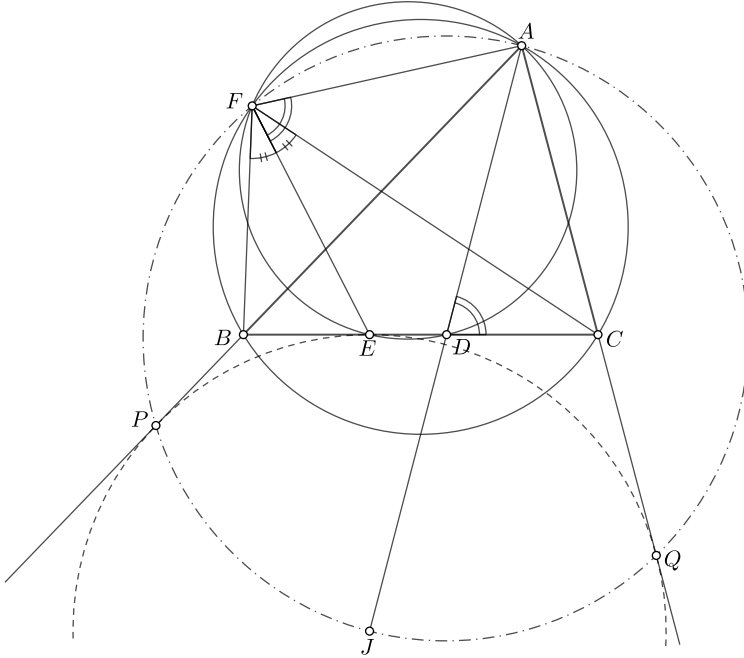
thus

$$\angle EFC = \angle EFA - \angle CFA = \angle BAD = \frac{1}{2} \angle BAC.$$

Because $\angle BFC = \angle BAC$, therefore line FE is angle bisector of BFC and we obtain that

$$\frac{BF}{CF} = \frac{BE}{CE} = \frac{BP}{CQ}.$$

According to this relation and $\angle FBP = 180^\circ - \angle FBA = 180^\circ - \angle FCA = \angle FCQ$ we may conclude that triangles FBP and FCQ are similar. It means that $\angle FPA = \angle FQA$. Hence $AFPQ$ is cyclic. Notice that AJ is its diameter, because $\angle APJ = \angle AQJ = 90^\circ$. It yields to conclusion that $\angle AFJ = 90^\circ$.



□

Problem 2. *Perpendicular bisector of side BC intersects circumcircle of triangle ABC in points P and Q . Points A and P lie on the same side of line BC . Point R is an orthogonal projection of point P on line AC . Point S is middle of line segment AQ . Show that points A, B, R, S lie on one circle.*

Proof. Without loss of generality let us assume that $AB < AC$. If $AB = AC$, then $A = P = R$ and $S = O$. Because in such situation $\angle ACB < 90^\circ$, therefore points A, B, O are not collinear, so they must lie on one circle and thesis of the problem is true. In case where $AB > AC$, we do similar reasoning.

Denote by O circumcenter of triangle ABC . It is obvious that segment PQ contains point O and $OS \perp AQ$. Since $\angle PCR = \angle OQS$, hence right-angled triangles PRC and OSQ are similar. Moreover $PB = PC$, $OB = OQ$ and

$$\angle BOQ = 2\angle BCQ = 2\angle QPC = \angle BPC,$$

so triangles OBQ and PBC are similar.

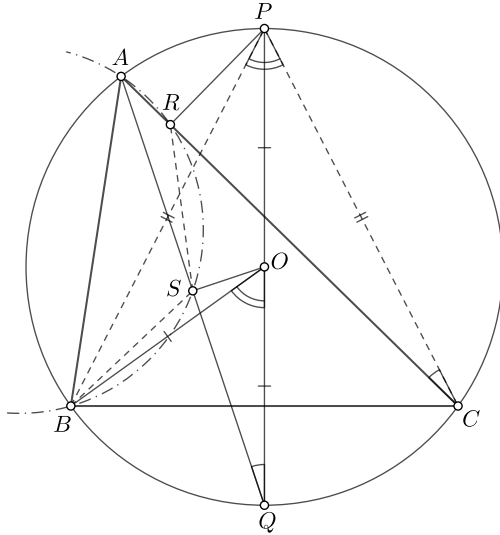
According to obtained similarities we may conclude that quadrilaterals $OSBQ$ and $PRBC$ are similar. In particular

$$\angle BSQ = \angle BRC,$$

so

$$\angle ARB = 180^\circ - \angle BRC = 180^\circ - \angle BSQ = \angle ASB,$$

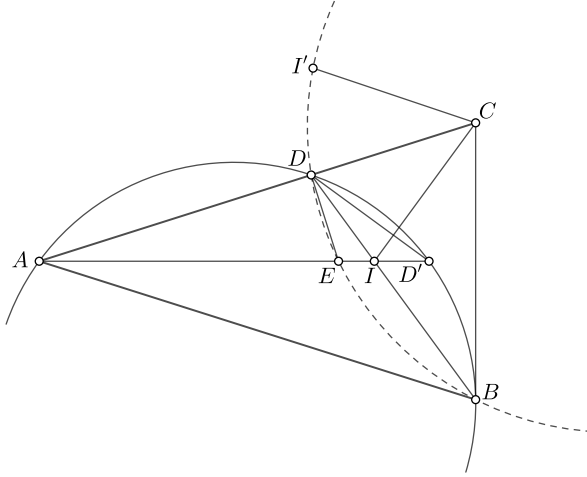
thus $ABSR$ is cyclic and we are done.



□

Problem 3. Let ABC be a triangle with $AB = AC \neq BC$ and let I be its incentre. The line BI meets AC at D , and the line through D perpendicular to AC meets AI at E . Prove that the reflection of I in AC lies on the circumcircle of triangle BDE .

Proof. Let I' be the reflection of I in AC , and let D' be the second intersection of AI and the circumcircle of triangle ABD . Since AD' bisects angle BAD , point D' is the midpoint of the arc BD and $DD' = BD' = CD'$. Obviously, A, E, D' lie on AI in this order.



We find that $\angle ED'D = \angle AD'D = \angle ABD = \angle IBC = \angle ICB$. Next, since D' is the circumcentre of triangle BCD , we have $\angle EDD' = 90^\circ - \angle D'DC = \angle CBD = \angle IBC$. The two relations show that triangles $ED'D$ and ICB are similar. Therefore, we have

$$\frac{BC}{CI'} = \frac{BC}{CI} = \frac{DD'}{D'E} = \frac{BD'}{D'E}.$$

Also, we get

$$\angle BCI' = \angle BCA + \angle ACI' = \angle BCA + \angle ICA = \angle BCA + \angle DBC = \angle BDA = \angle BD'E.$$

These show triangles BCI' and $BD'E$ are similar, and hence triangles BCD' and $BI'E$ are similar. As BCD' is isosceles, we obtain $BE = I'E$.

As DE is the external angle bisector of angle BDI' and $EI' = EB$, we know that E lies on the circumcircle of triangle BDI' . $\square$

Problem 4. Consider the parallelogram $ABCD$ with obtuse angle at A . Let H be the feet of perpendicular from A to the side BC . The median from C in triangle ABC meets the circumcircle of triangle ABC at the point K . Prove that points $KHCD$ is cyclic.

Proof. Let M be the midpoint of segment AB . Note that M is the circumcenter of the right-angled triangle ABH . Thus

$$\begin{aligned}\angle KMH &= \angle KMB + \angle BMH = (\angle MBC + \angle MCB) + 2\angle BAH = \\ &= \angle MBC + \angle MCB + 180^\circ - 2\angle ABH = \angle MCB + 180^\circ - \angle ABH = \\ &= \angle KAB + \angle BAD = \angle KAD.\end{aligned}$$

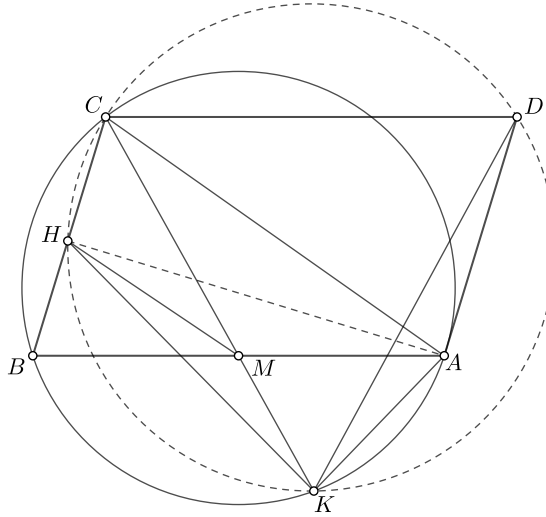
Moreover, since $\triangle KAM \sim \triangle BCM$ (from SSS), we obtain that

$$\frac{KM}{KA} = \frac{BM}{BC} = \frac{MH}{AD}$$

So $\triangle KMH \sim \triangle KAD$. Therefore

$$\angle KDC = \angle ADC - \angle ADK = \angle ABC - \angle MHK = \angle MHB - \angle MHK = 180^\circ - \angle KHC.$$

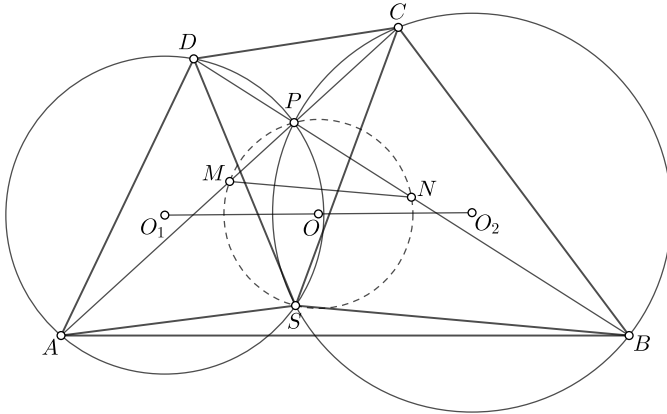
Hence $KHCD$ is cyclic and we are done.



□

Problem 5. Let $ABCD$ be a quadrilateral. Let diagonals AC and BD meet at P . Let O_1 and O_2 be the circumcenters of APD and BPC . Let M , N , and O be the midpoints of AC , BD , and O_1O_2 . Show that O is the circumcenter of MPN .

Proof. Denote by S the second intersection on these two circles (denote them by ω_1 and ω_2 , respectively). Notice that triangle SAD is similar to triangle SCB . Therefore S is the center of spiral similarity that sends A to C , D to B , ω_1 to ω_2 and thus also O_1 to O_2 .



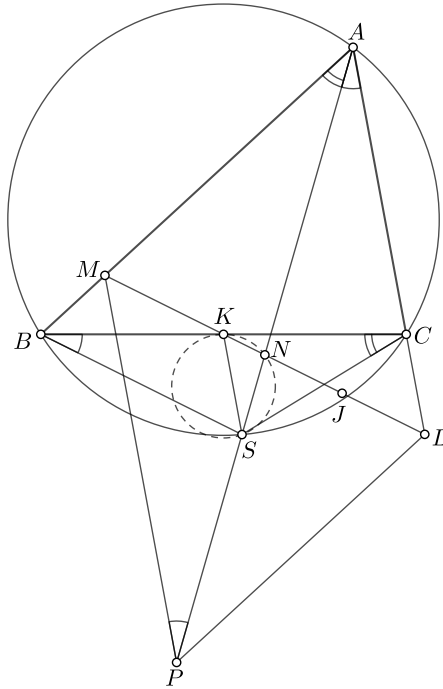
As triangle SAD glides to triangle SCB , its circumcenter O_1 glides along O_1O_2 and since P and S are symmetric with respect to O_1O_2 , its circumcircle at all time passes through P . Focusing on the situation in the middle of its way we realize that S , M , N and P lie on a circle with center O . Hence we are done. $\square$

Problem 6. In triangle ABC denote by K the midpoint of BC . Let M be a point of the segment AB . Line KM intersects AC at L , such that C lies between A and L . Point N is a midpoint of LM . Line AN intersects circumcircle of ABC at $S \neq A$. Prove that if $S \neq N$, then circle passing through K, N and S is tangent to BC .

Proof. First we will show that N lies between point K and L . Denote by J symmetric point to point M with respect to point K . Triangles KBM and KCJ are congruent, thus

$$\angle ACJ = \angle ACB + \angle KCJ = \angle ACB + \angle KBM < 180^\circ.$$

It means that J lies between K and L , so $KL > KJ = KM$ and thus midpoint N of segment ML lies on segment KL . Therefore N and A lie on opposite sides of BC (we cannot forget that order of points A, N, S on line AN can differ).



Let P be symmetric point to point A about point N . Then $AMPL$ is a parallelogram. Thus

$$\angle SCB = \angle SAB = \angle PAM$$

and

$$\angle SBC = \angle SAC = \angle APM.$$

Therefore triangles BSC and PMA are similar. Consider similarity, which transforms one triangle into the other. Then it maps midpoint K of BC to midpoint N

of PA . Hence triangles SKC and MNA are similar too. It means that

$$\sphericalangle SKC = \sphericalangle MNA = \sphericalangle KNA = \begin{cases} 180^\circ - \sphericalangle KNS & \text{if points are in order } A, N, S, \\ \sphericalangle KNS & \text{if points are in order } A, S, N. \end{cases}$$

Consider circumcircle of triangle KNS . Points A and C lie on the same side of line KN . It means that no matter in which order lie points A, N, S , obtained equality always implies that line AC is tangent to circumcircle of triangle KNS . $\square$

Problem 7.*Proof.*

□

Problem 8.*Proof.*

□

Problem 9.*Proof.*

□

Problem 10.*Proof.*

□

Problem 11.*Proof.*

□

Problem 12.*Proof.*

□

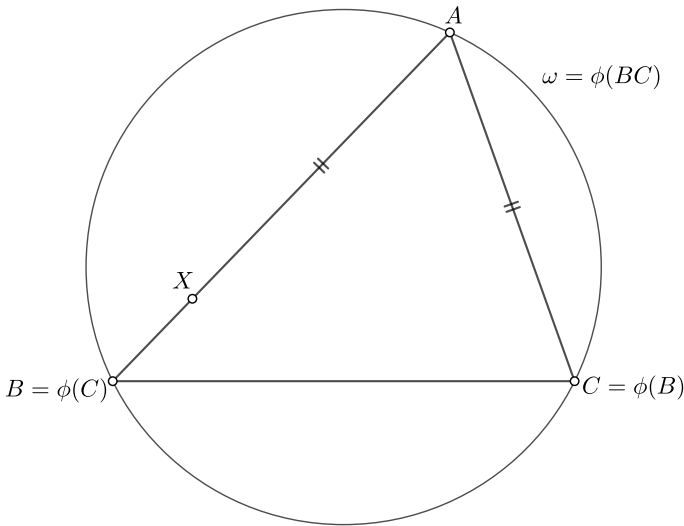
2.4. Inversion $\sqrt{bc}$.

Problem 1. *Given is a triangle ABC . Consider such transformation ϕ which is composition of inversion with respect to circle with center A and radius $\sqrt{AB \cdot AC}$ and symmetry about the angle bisector of angle BAC . Show that*

(a) $\phi(B) = C$, $\phi(C) = B$;

(b) $\phi(\omega) = BC$, $\phi(BC) = \omega$, where ω is circumcircle of triangle ABC .

Proof. (a) Notice that inversion centred at point A with radius $\sqrt{AB \cdot AC}$ maps point B to point X on line AC such that $AX = AC$. Thus after symmetry with respect to angle bisector of BAC , X will be mapped to point C . Hence $\phi(B) = C$. Similarly we obtain that $\phi(C) = B$.



(b) Inversion centred at point A with radius $\sqrt{AB \cdot AC}$ transforms line BC into circle passing through point A . Since $\phi(B) = C$ and $\phi(C) = B$, thus $\phi(BC) = \omega$. Similarly $\phi(\omega) = BC$. $\square$

Problem 2. Given is a triangle ABC . Consider such transformation ϕ which is composition of inversion with respect to circle with center A and radius $\sqrt{\frac{1}{2}AB \cdot AC}$ and symmetry about the angle bisector of angle BAC . Prove that

(a) $\phi(O) = H_A$, where O is circumcenter of ABC , and H_A is the base of altitude from vertex A ;

(b) circumcircle of triangle BOC is mapped to nine point circle of ABC .

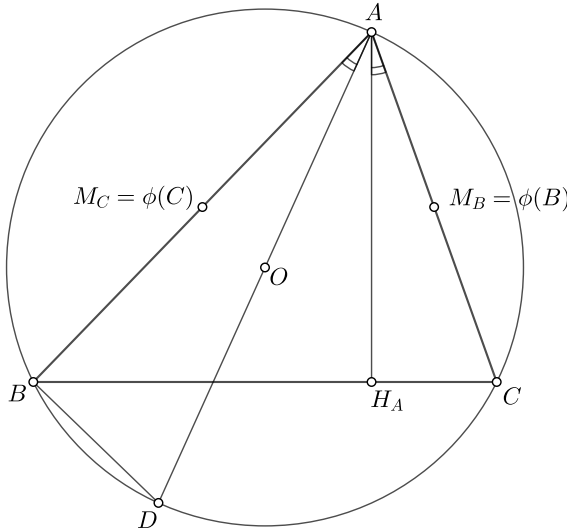
Proof. (a) It is well-known that AO and AH_A are isogonal conjugated, which means they are symmetric with respect to angle bisector of angle BAC . Hence we need to show that

$$AH_A \cdot AO = \frac{1}{2}AB \cdot AC.$$

Denote by D the intersection of AO with circumcircle of triangle ABC . Then triangles AH_AB and ACD are similar, since $\angle AH_AB = \angle ACD = 90^\circ$ and $\angle BAH_A = \angle DAC$ from isogonality. Therefore

$$\frac{AB}{AH_A} = \frac{2AO}{AC} \implies AH_A \cdot AO = \frac{1}{2}AB \cdot AC,$$

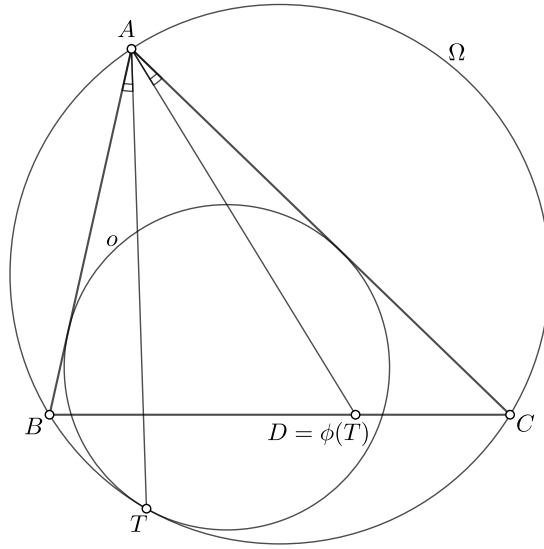
so we are done.



(b) Denote by M_B and M_C midpoints of segments AC and AB , respectively. According to previous problem, it's easy to notice that $\phi(B) = M_B$ and $\phi(C) = M_C$. But since $\phi(O) = H_A$, thus circumcircle of triangle BOC is mapped to nine point circle of ABC . $\square$

Problem 3. Let Ω be circumcircle of triangle ABC . Denote by o circle tangent to AB , AC and internally tangent to ω at point T . Point D is tangency point of A -excircle with line BC . Prove that $\sphericalangle BAT = \sphericalangle DAC$.

Proof. Consider such transformation ϕ which is composition of inversion with respect to circle with center A and radius $\sqrt{AB \cdot AC}$ and symmetry about the angle bisector of angle BAC . Denote by ω A -excircle of triangle ABC . It is sufficient to show that $\phi(o) = \omega$. But since $\phi(AB) = AC$, $\phi(AC) = AB$, $\phi(\Omega) = BC$ and o is tangent to AB , AC and Ω , thus after transformation ϕ circle o is mapped into A -excircle of triangle ABC . In particular $\phi(\sphericalangle BAT) = \sphericalangle DAC$, hence $\sphericalangle BAT = \sphericalangle DAC$ and we are done.



□

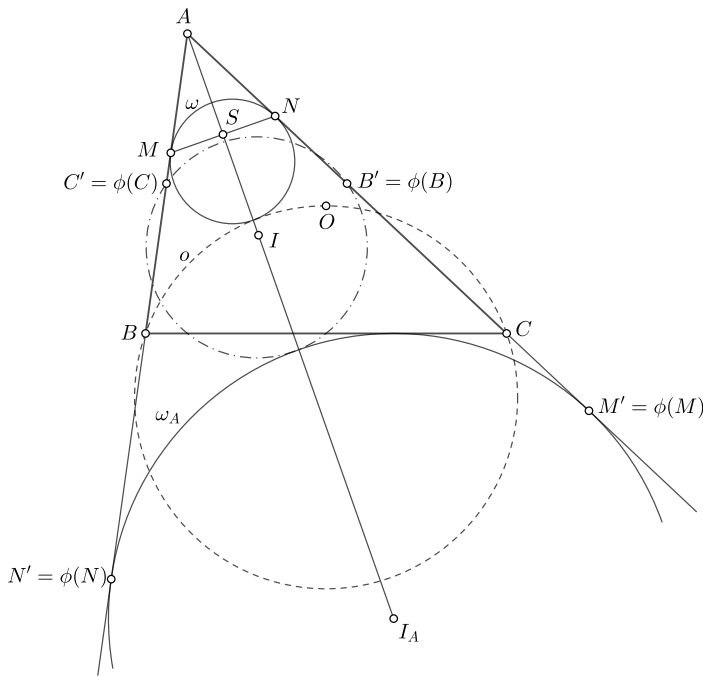
Problem 4. Denote by O and I circumcenter and incenter of triangle ABC , respectively. Let circle ω tangent to AB , AC and circumcircle o of triangle BOC be tangent to AB and AC at points M and N , respectively. Show that MN bisects AI .

Proof. Consider such transformation ϕ which is composition of inversion with respect to circle with center A and radius $\sqrt{\frac{1}{2}AB \cdot AC}$ and symmetry about the angle bisector of angle BAC .

Notice that $\phi(B)$ and $\phi(C)$ are midpoints of segments AC and AB , respectively. Denote by H_A base of altitude from vertex A . Recall a well-known formula $A = \frac{abc}{4R}$ for area of triangle. Notice that

$$AO \cdot AH_A = \frac{AB \cdot BC \cdot CA}{4[ABC]} \cdot \frac{2[ABC]}{BC} = \frac{1}{2}AB \cdot AC,$$

where $[F]$ stands for area of figure F . It means that $\phi(O) = H_A$. Therefore $\phi(o) = \omega_E$ is nine-point circle of triangle ABC .



Let ω_A be A -excircle of triangle ABC . According to well-known Feuerbach theorem ² ω_A and ω_E are externally tangent. Note that $\phi(\omega) = \omega_A$, because ω after ϕ must be tangent to AB , AC and BC (BC is circumcircle after ϕ). Therefore points $M' = \phi(M)$ and $N' = \phi(N)$ are tangency points of ω_A to lines AC , AB ,

²**Feuerbach's theorem** states that the nine-point circle of any triangle is tangent internally to the incircle and tangent externally to the three excircles.

respectively. Thus

$$AM \cdot AM' = AN \cdot AN' = \frac{1}{2}AB \cdot AC.$$

Let S be midpoint of segment MN . Notice that triangles AMS and $AI_A M'$ are similar, where I_A is A -excenter. Hence

$$AS = \frac{AM' \cdot AM}{AI_A} = \frac{\frac{1}{2}AB \cdot AC}{AI_A}.$$

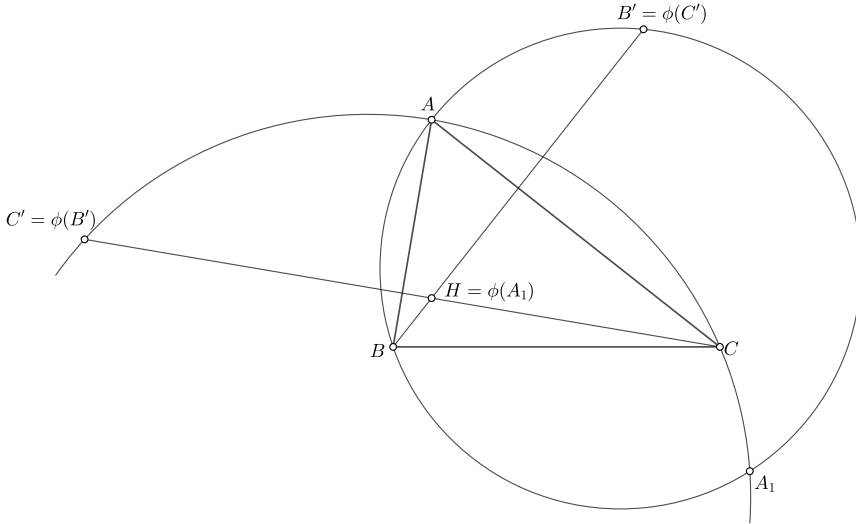
Now consider such transformation ϕ' which is composition of inversion with respect to circle with center A and radius $\sqrt{AB \cdot AC}$ and symmetry about the angle bisector of angle BAC . It is well-known that in such transformation incircle is sent to ω_A , thus $AI \cdot AI_A = AB \cdot AC$. It means that

$$\frac{AS}{AI} = \frac{\frac{\frac{1}{2}AB \cdot AC}{AI_A}}{\frac{AB \cdot AC}{AI_A}} = \frac{1}{2}.$$

Hence $AI = 2AS$ and $AS = SI$, so we are done. □

Problem 5. Let ABC be a triangle and A' , B' , C' the symmetric of vertex about opposite sides. The intersection of the circumcircles of triangles ABB' and ACC' is A_1 . Points B_1 and C_1 are defined similarly. Prove that lines AA_1 , BB_1 and CC_1 are concurrent.

Proof. Let ϕ refer to the inversion around A with radius $\sqrt{AB \cdot AC}$ followed by reflection across the angle bisector of A .



Under ϕ , A_1 gets sent to the H , the orthocenter of triangle ABC (because it is easy to check that $\phi(C) = B$ and $\phi(C') = B'$, thus circumcircle of triangle ACC' is sent to B -altitude; analogously circumcircle of ABB' is being sent to C -altitude and we may conclude that $\phi(A_1) = H$). Thus, AA_1 is isogonal to the A -altitude. Similar assumption holds for BB_1 and CC_1 . Thus, they are concurrent at the isogonal conjugate of the orthocenter (namely O). $\square$

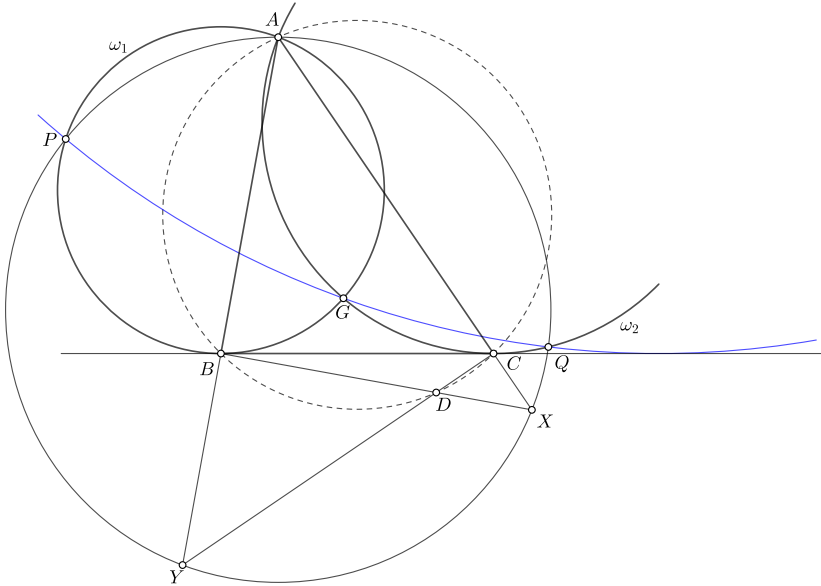
Problem 6. Circles ω_1 and ω_2 intersect at X and Y and are tangent internally to circle Ω , respectively at points A and B . Line AB intersect again circles ω_1 and ω_2 at A_1 , B_1 , respectively. Circle inscribed in curvilinear triangle A_1B_1X is touches A_1B_1 at the point Z . Prove that $\angle AXZ = \angle BXZ$.

Proof. napsac solv

□

Problem 7. Consider point D on circumcircle Ω of ABC such that AD is diameter of this circle. Let DB, DC intersect AC, AB at X, Y , respectively. ω_1 is circle passing through A, B and tangent to BC ; analogously ω_2 passes through A, C and is tangent to BC . Let ω_1 and ω_2 intersect at G ; circumcircle of AXY intersect ω_1, ω_2 at P, Q , respectively. Prove that circumcircle of triangle GPQ is tangents to BC .

Proof. Invert about A with radius $\sqrt{AB \cdot AC}$ followed by reflection in the angle bisector of angle BAC . It's not hard to notice that $\phi(\omega_1) = \ell_C$, where ℓ_C stands for tangent to Ω at point C . Analogously $\phi(\omega_2) = \ell_B$, where ℓ_B stands for tangent to Ω at point B . If $\ell_B \cap \ell_C = K$, then $\phi(G)$ is obviously K .



Denote by H_A, H_B, H_C bases of altitudes in triangle ABC from vertexes A, B, C , respectively. It is clear that $\phi(D) = H_A$ and therefore by basic properties of inversion $\sqrt{bc}$ we may conclude that $\phi(X) = H_C$ and $\phi(Y) = H_B$. Hence circumcircle of triangle AXY is sent to line $H_B H_C$. If $\ell_B \cap H_B H_C = Q', \ell_C \cap H_B H_C = P'$, then $\phi(P) = P'$ and $\phi(Q) = Q'$.

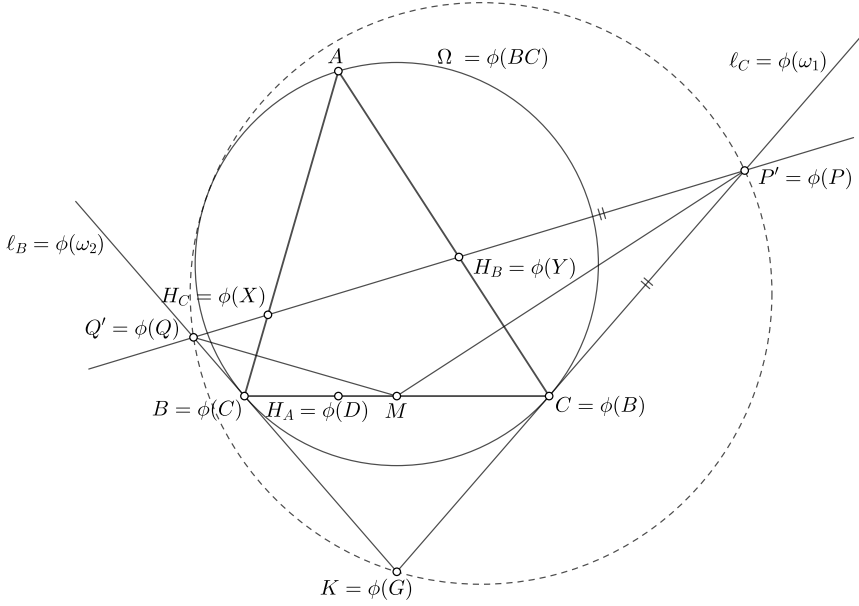
Since $\phi(BC) = \Omega$, thus we want to prove that circumcircle of triangle $KP'Q'$ is tangent to Ω .

Let M be the segment BC . Then we have $MH_B = MC$, which means that M lies on the perpendicular bisector of CH_B . Moreover

$$\sphericalangle P'CH_B = \sphericalangle ABC = \sphericalangle AH_B H_C = \sphericalangle P'H_B C \implies P'C = P'H_B.$$

This means that $P'M$ is the perpendicular bisector of CH_B , or equivalently that PM is the internal angle bisector of angle $Q'P'K$. Similarly, we can show that QM is the internal angle bisector of angle $P'Q'K$. It means that M is in fact the incenter of triangle $KP'Q'$.

Since $KM \perp BC$, thus points B and C must be the points where the K -mixtilinear incircle ³ of $KP'Q'$ touches KP' and KQ' (by Sawayama Theorem). However, such a circle is Ω only, and so Ω is the K -mixtilinear incircle of KPQ , which is tangent to circumcircle of $KP'Q'$ by definition. Hence we are done.



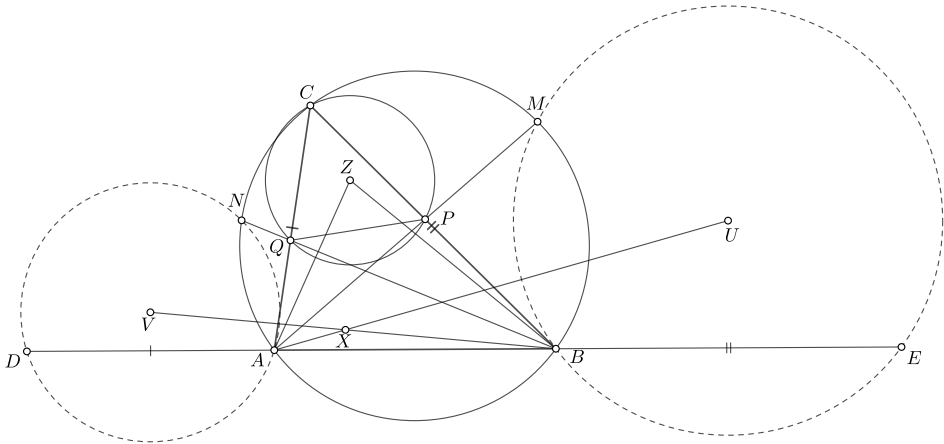
□

³A circle that is internally tangent to two sides of a triangle and to the circumcircle is called a mixtilinear incircle. There are three mixtilinear incircles, one corresponding to each angle of the triangle.

Problem 8. Given is triangle ABC . Points D and E lie on line AB in order D, A, B, E . Moreover $AD = AC$ and $BE = BC$. Angle bisector of angles at BAC and ABC intersect BC , AC at P and Q , and circumcircle of ABC at M and N , respectively. Line connecting A with circumcenter of BME and line connecting B with circumcenter of AND intersect at X . Prove that $CX \perp PQ$.

Proof. Denote by U circumcenter of triangle BME . Apply transformation ϕ which consist of inversion about A with radius $\sqrt{AB \cdot AC}$ followed by reflection in the angle bisector of angle BAC . Then $\phi(B) = C$, $\phi(C) = B$, $\phi(M) = P$. Notice that $\phi(E) = Q$, because

Thus circumcircle of triangle BME is sent to circumcircle of triangle CPQ . Denote by Z circumcenter of triangle CPQ . We may now conclude AU and AZ are symmetric with respect to AP (angle bisector of angle BAC).



By similar reasoning we obtain that lines BZ and BV , where V stands for circumcenter of triangle DAN , are symmetric in BQ . Notice that according to obtained symmetries we may conclude that points X and Z are isogonal conjugated. Thus lines AX and AZ are symmetric with respect to angle bisector of ACB (in fact they are isogonal conjugated). Recall a well-known fact that orthocenter and circumcenter are isogonal conjugated. Thus, since Z is circumcenter of triangle CPQ , line AX must contain the A -altitude of triangle CPQ . Hence $AX \perp PQ$ and we are done. $\square$